SAE International™

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(R) Data Link Layer

RATIONALE

This SAE Recommended Practice revision incorporates changes to the abort reasons for transport protocol sessions. Appendix C, Figure C7, updated to correct formatting and in right side of figure change TP.DT.2 to TP.DT.3. Added the request response guidance in section 5.4.2, 5.4.6 and 5.10.3.2. Change SPN 2554 name to Controller Application Identity.

FOREWORD

This series of SAE Recommended Practices have been developed by the Truck and Bus Control and Communications Network Committee of the Truck and Bus Electrical and Electronics Steering Committee. The objectives of the committee are to develop information reports, recommended practices, and standards concerned with the requirements, design, and usage of devices that transmit electronic signals and control information among vehicle components. The usage of these Recommended Practices is not limited to truck and bus applications; other applications may be accommodated with immediate support being provided for construction and agricultural equipment, and stationary power systems.

These SAE Recommended Practices are intended as a guide toward standard practice and are subject to change so as to keep pace with experience and technical advances.

This data link layer is used for all SAE J1939 applications. It is the characteristic that is common across all SAE J1939 applications.

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1. SCOPE

The SAE J1939 documents are intended for light, medium, and heavy-duty vehicles used on or off road as well as appropriate stationary applications which use vehicle derived components (e.g. generator sets). Vehicles of interest include, but are not limited to, on- and off-highway trucks and their trailers, construction equipment, and agricultural equipment and implements.

The purpose of these documents is to provide an open interconnect system for electronic systems. It is the intention of these documents to allow Electronic Control Units to communicate with each other by providing a standard architecture.

This particular document, SAE J1939-21, describes the data link layer using the CAN protocol with 29-bit Identifiers. For SAE J1939 no alternative data link layers are permitted.

2. REFERENCES

2.1 Applicable Documents

General information regarding this series of recommended practices is found in SAE J1939. The latest issue of SAE publications shall apply.

2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or 724-776-4970 (outside USA), www.sae.org.

SAE J1939 Recommended Practice for a Serial Control and Communications Vehicle Network is the parent document and should be referenced in general.
SAE J1939-71 Vehicle Application Layer
SAE J1939-81 Network Management

2.2 Related Publications

The following publications are provided for information purposes only and are not a required part of this SAE Technical Report.

2.2.1 ISO Publications

Available from American National Standards Institute, 25 West 43rd Street, New York, NY 10036-8002, Tel: 212-642-4900, www.ansi.org.

ISO 11898 (Amended) Road Vehicles—Interchange of digital information—Controller area network (CAN) for high-speed communications AMENDMENT 1

ISO 15765-3 Road Vehicles—Diagnostics on Controller Area Network (CAN)—Part 3; Implementation of Unified Diagnostic Services (UDS on CAN), October 23, 2003

3. DEFINITIONS

Terms and definitions are defined in SAE J1939.

4. ABBREVIATIONS

ACK Acknowledgment
BAM Broadcast Announce Message
CAN Controller Area Network
CRC Cyclic Redundancy Check
CTS Clear-to-Send

DA	Destination Address
DLC	Data Length Code
DP	Data Page
DT	Data Transfer
EDP	Extended Data Page
EOF	End of Frame
GE	Group Extension
ID	Identifier
IDE	Identifier Extension Bit
LLC	Logical Link Control
LSB	Least Significant Byte or Least Significant Bit
MAC	Medium Access Control
MF	Manufacturer
MSB	Most Significant Byte or Most Significant Bit
NA	Not Allowed
NACK	Negative-Acknowledgment
P	Priority
PDU	Protocol Data Unit
PF	PDU Format
PGN	Parameter Group Number
PS	PDU Specific
RTR	Remote Transmission Request
RTS	Request-to-Send
SA	Source Address
SOF	Start of Frame
SPN	Suspect Parameter Number
SRR	Substitute Remote Request
TP	Transport Protocol
T_h	Hold Time
T_r	Response Time
un	Undefined

5. TECHNICAL REQUIREMENTS

The data link layer provides for the reliable transfer of data across the physical link. This consists of sending the CAN Data Frame with the necessary synchronization, sequence control, error control, and flow control. The flow control is accomplished by a consistent message/frame format.

5.1 Message/Frame Format

Message format conforms to the CAN requirements. The CAN specification referenced throughout this document is *CAN Specification 2.0 Part B*, September 1991. It should be noted that when there are differences between the CAN specification and SAE J1939, then SAE J1939 is the guiding document.

The CAN document specifies, in an information routing related discussion, that node addresses are not used. While this is true for some applications of CAN, it is not true for SAE J1939. The definition of the SAE J1939 network requires that node addressing be used to prevent multiple nodes from using the same CAN Identifier field (see SAE J1939). Many additional requirements exist in SAE J1939 that are not specified by CAN.

CAN 2.0B contains specification of two message formats, standard frame and extended frame.

CAN 2.0B compatibility implies that messages of both formats can potentially be present on a single network by using certain bit codings that allow for the recognition of the different formats. To this point, SAE J1939 also has accommodations for both CAN data frame formats. But SAE J1939 only defines a full strategy for standardized communications using the extended frame format. All standard frame format messages are for proprietary use following the rules defined in this document.

Therefore, SAE J1939 devices MUST use the extended frame format. Standard frame format messages can reside on the network, but only as described in this document.

Note—Standard frame devices do not respond to network management messages and are not able to support the strategy for standardized communications.

Referencing Figure 1, the CAN data frame is parsed into different bit fields. The number and parsing of the bits in the arbitration and control fields differ between the CAN standard and extended frame messages. CAN standard frame messages, shown in "A," contains 11 Identifier bits in the arbitration field and CAN extended frame messages, shown in "B," contain 29 identifier bits in the arbitration field. SAE J1939 has further defined the identifier bits in the arbitration field of the CAN Data Frame formats. These definitions can be seen in Table 1.

5.1.1 SAE J1939 Message Frame Format ("CAN 2.0B" Extended Frame Format)

The CAN extended frame message, as shown in Figure 1, encompasses a single Protocol Data Unit (PDU). PDUs consist of seven predefined fields. These fields are assimilated from information provided by the application layer. They are priority, extended data page, data page, PDU format, PDU specific (which can be a destination address, group extension, or proprietary), source address and data fields. They are then packaged into one or more CAN Data Frames and sent over the physical media to other network devices. The layers of the Open Systems Interconnect (OSI) model that SAE J1939 supports are shown in Figure 2. It should be recognized that some Parameter Group definitions require more than one CAN Data Frame to send their information.

Table 1 shows the arbitration and control fields of the following: 29-bit identifier for CAN; 29-bit identifier for SAE J1939; 11-bit identifier for CAN; and the 11-bit identifier for SAE J1939. A complete definition for each of the SAE J1939 bit field assignments can be found in the sections that define the SAE J1939 Protocol Data Unit (see Section 5.2). Throughout this document the CAN Data Frame data field is discussed as bytes 1 through 8. Byte 1's most significant bit (bit 8) is the first bit sent closest to the DLC field. Byte 8's least significant bit (bit 1) is last of the data bits to be sent and is closest to the CRC Field.

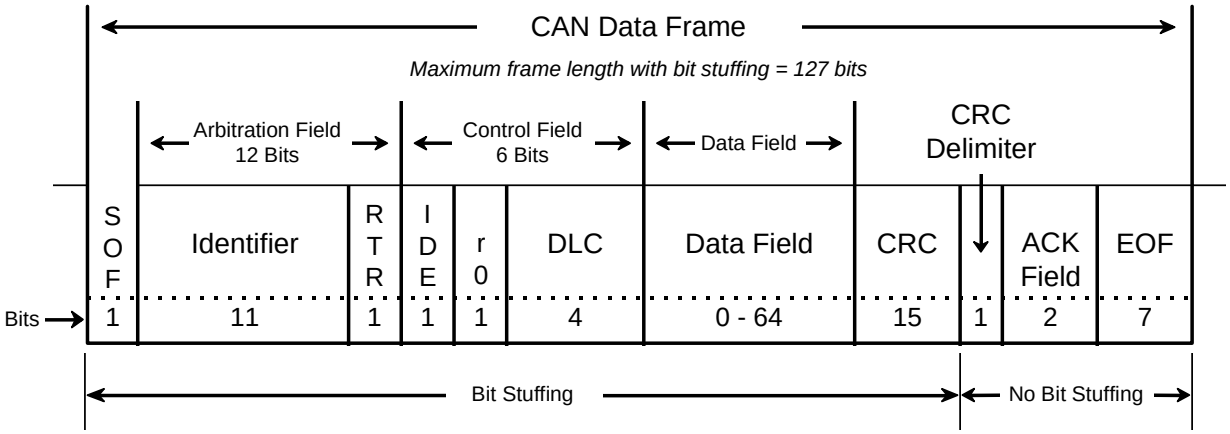
When the Extended Data Page (EDP) is equal to 1 and the Data Page (DP) is equal to 1, this identifies the CAN frame as an ISO 15765-3 formatted frame. Therefore, the processing of the CAN frame will not follow the definitions provided by J1939 specifications. The user should consult the ISO 15765-3 specifications for further guidance.

For the case where the Extended Data Page is equal to 1 and the Data Page is equal to 0, the SAE J1939 committee has not specified that the CAN identifier will be utilized the way it is specified in the remaining sections of this document. Therefore, those CAN frames may require unique handling. This will be specified in future versions of the J1939-21 document.

5.1.2 Parameter Group Number (PGN)

Whenever it is necessary to identify a Parameter Group in the data field of a CAN data frame, it will be expressed in 24 bits. The 24-bit value is sent least significant byte first (see Table 2 where it is stated the MSB is sent third, middle byte second and least significant byte first). The Parameter Group Number is a 24-bit value that has the following constituent components: 6 bits set to zero, Extended Data Page bit, Data Page bit, PDU Format Field (8 bits), and Group Extension Field (8 bits). The procedure for the bit fields to be converted to a Parameter Group Number is as follows. The six most significant bits of the PGN are set to zero. Then the Extended Data Page Bit, Data Page bit and PDU Format field are copied into the next 10 bits. Lastly if the PF value is less than 240 ($F0_{16}$), then the least significant byte of the PGN is set to zero. Otherwise it is set to the value of the Group Extension Field. Reference Table 2 for an illustration of the constituent component bit fields and their corresponding PGN decimal number. Note that not all 2^{17} (or 131072) combinations are available for assignment using the conventions specified in this document. The conventions specified yield 8672 PGs (calculated as: 2 pages * [240 + (16*256)] = 8672 PGs). The Cumulative Number of PGs column in Table 2 identifies the number of PGNs possible. See SAE J1939, Appendix A to see the assigned PGNs.

A. CAN Standard Frame Format



B. CAN Extended Frame Format

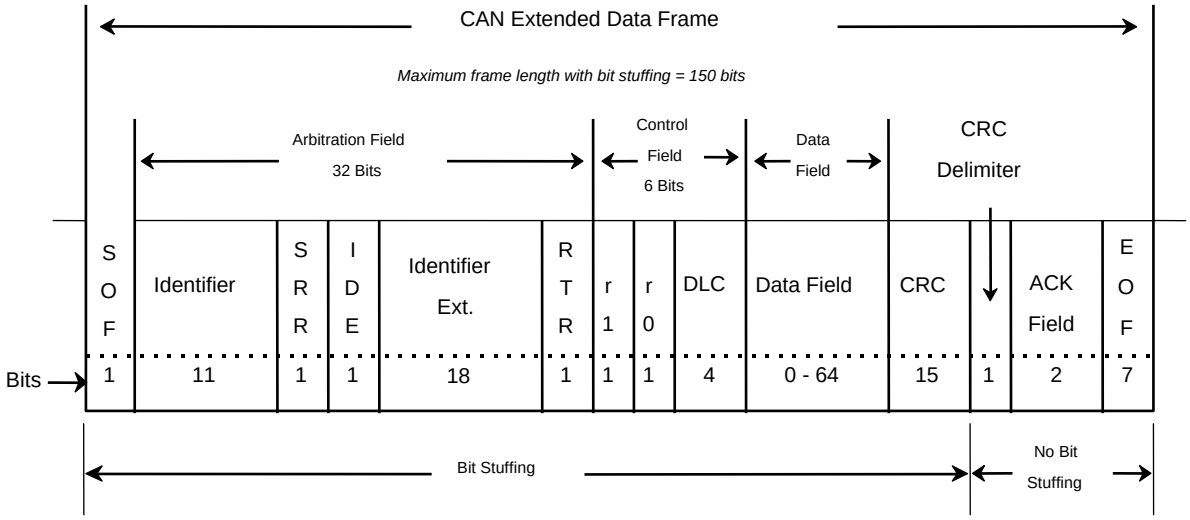


FIGURE 1 - CAN DATA FRAMES

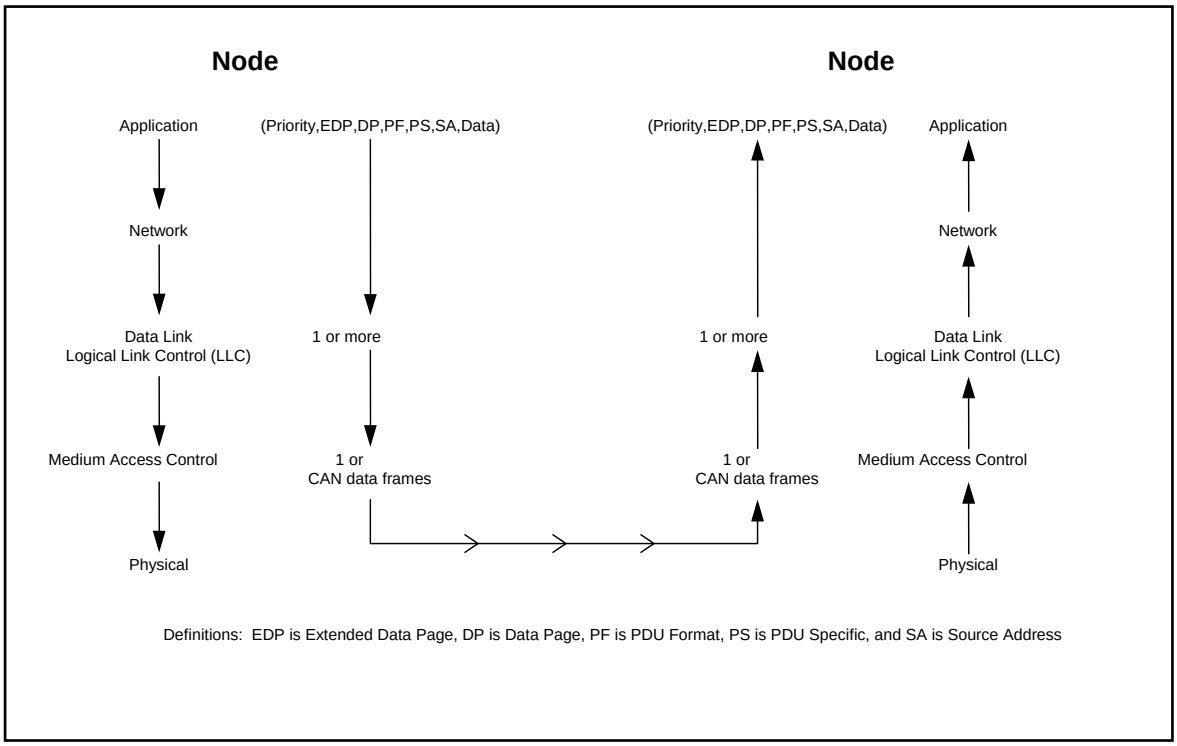


FIGURE 2 - APPLICATION OF OSI MODEL BY SAE J1939

TABLE 1 - MAPPING OF SAE J1939 INTO CAN'S ARBITRATION AND CONTROL FIELDS

29-Bit Identifiers CAN	29-Bit Identifiers SAE J1939	Frame Bit Position	11-Bit Identifiers CAN	11-Bit Identifiers SAE J1939[1]
SOF	SOF*	1	SOF	SOF*
ID 28	P 3	2	ID 11	P 3
ID 27	P 2	3	ID 10	P 2
ID 26	P 1	4	ID 9	P 1
ID 25	EDP	5	ID 8	SA 8
ID 24	DP	6	ID 7	SA 7
ID 23	PF 8	7	ID 6	SA 6
ID 22	PF 7	8	ID 5	SA 5
ID 21	PF 6	9	ID 4	SA 4
ID 20	PF 5	10	ID 3	SA 3
ID 19	PF 4	11	ID 2	SA 2
ID 18	PF 3	12	ID 1	SA 1
SRR (r)	SRR*	13	RTR (x)	RTR* (d)
IDE (r)	IDE*	14	IDE (d)	IDE*
ID 17	PF 2	15	r 0	r 0*
ID 16	PF 1	16	DLC 4	DLC 4
ID 15	PS 8	17	DLC 3	DLC 3
ID 14	PS 7	18	DLC 2	DLC 2
ID 13	PS 6	19	DLC 1	DLC 1
ID 12	PS 5	20		
ID 11	PS 4	21		
ID 10	PS 3	22		
ID 9	PS 2	23		
ID 8	PS 1	24		
ID 7	SA 8	25		
ID 6	SA 7	26		
ID 5	SA 6	27		
ID 4	SA 5	28		
ID 3	SA 4	29		
ID 2	SA 3	30		
ID 1	SA 2	31		
ID 0	SA 1	32		
RTR (x)	RTR* (d)	33		
r 1	r 1*	34		
r 0	r 0*	35		
DLC 4	DLC 4	36		
DLC 3	DLC 3	37		
DLC 2	DLC 2	38		
DLC 1	DLC 1	39		

NOTES

SOF - Start of Frame Bit
ID## - Identifier Bit #n
SRR - Substitute Remote Request
RTR - Remote Transmission Request Bit
IDE - Identifier Extension Bit
r# - CAN Reserved Bit #n
DLC# - Data Length Code Bit #n
(d) - dominant bit
(r) - recessive bit

P# - SAE J1939 Priority Bit #n
EDP - SAE J1939 Extended Data Page
SA# - SAE J1939 Source Address Bit #n
DP - SAE J1939 Data Page
PF# - SAE J1939 PDU Format Bit #n
PS# - SAE J1939 PDU Specific Bit #n
[1] Required format of proprietary 11-bit identifiers
*CAN Defined Bit, Unchanged in SAE J1939
(x) - bit state dependent on message

TABLE 2 - PARAMETER GROUP NUMBER EXAMPLES (SAE USE EDP=0)

PGN Constituent Components	PGN Constituent Components	PGN Constituent Components	PGN Constituent Components	PGN Constituent Components					
PGN (MSB)	PGN (MSB)	PGN (MSB)	PGN	PGN (LSB)					
Byte 1	Byte 1	Byte 1	Byte 2	Byte 3					
Sent third in CAN Data Frame			Sent second in CAN Data Frame	Sent first in CAN Data Frame					
	EDP	DP	PF	PS	PGN	PGN	Number of Assignable PGs	Cumulative Number of PGs	SAE or Manufacturer Assigned
Bits 8-3	Bit 2	Bit 1	Bits 8-1	Bits 8-1	Dec_{10}	Hex_{16}			
0	0	0	0	0	0	000000_{16}			SAE
							239	239	
0	0	0	238	0	60928	$00EE00_{16}$			
0	0	0	239	0	61184	$00EF00_{16}$	1	240	MF
0	0	0	240	0	61440	$00F000_{16}$			SAE
							3840		
0	0	0	254	255	65279	$00FEFF_{16}$		4080	
0	0	0	255	0	65280	$00FF00_{16}$			
							256		MF
0	0	0	255	255	65535	$00FFFF_{16}$		4336	
0	0	1	0	0	65536	010000_{16}			
							239	4575	SAE
0	0	1	238	0	126464	$01EE00_{16}$			
0	0	1	239	0	126720	$01EF00_{16}$	1	4576	MF
0	0	1	240	0	126976	$01F000_{16}$			
							4096		SAE
0	0	1	255	255	131071	$01FFFF_{16}$		8672	
TOTALS							8672	8672	

5.1.3 SAE J1939 Support of “CAN 2.0B” Standard Frame Format Messages

It is recognized that controllers on the SAE J1939 network may support the CAN standard frame (11-bit identifier) message format. Though these are not compatible with the SAE J1939 message structure, to accommodate the coexistence of the two formats, a minimum level of definition is given. This minimum definition allows devices that use this format to not interfere with other devices. CAN standard frame format messages are defined to be proprietary. Referencing Table 1, the 11-bit identifier field is parsed as follows: the three most significant bits are used as priority bits; the eight least significant bits identify the source address of the PDU. Priority bits are described in Section 5.2.1. The source address is defined in the Source Address Table (see SAE J1939).

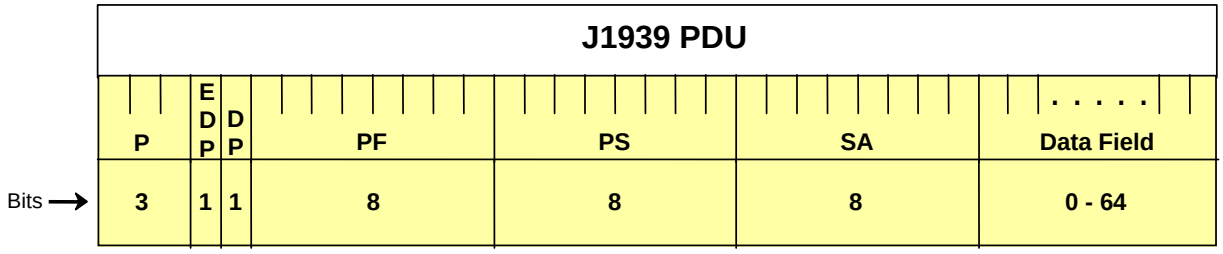
NOTE—Incorrect bus arbitration could occur when two messages, one standard frame and one extended frame, attempt to access the bus at the same time. The source address (SA) has a higher relative priority in the standard frame messages than in the extended frame messages. The message with the 11-bit identifier (standard frame) could have an SA indicating a higher priority than the extended data page bit, data page bit and PF of the 29-bit identifier (extended frame) message. The three priority bits should be used to achieve the correct bus arbitration.

SAE J1939 only defines a full strategy for standardized communications using the extended frame format. Hardware conforming to CAN 2.0A specification MUST NOT be used on the network since these versions of hardware do not allow the extended frame messages to be communicated.

5.2 Protocol Data Unit (PDU)

The applications and/or network layer provide a string of information that is assimilated into a protocol data unit. The protocol data unit provides a framework for organizing the information that is key to each CAN Data Frame that is sent. The SAE J1939 protocol data unit consists of seven fields. They are priority, extended data page, data page, PDU format, PDU specific (which can be a destination address, group extension, or proprietary), source address, and data fields. They are then packaged into one or more CAN data frames and sent over the physical media to other network devices. There is only one PDU per CAN data frame. It should be recognized that some Parameter Group Number definitions require more than one CAN Data Frame to send the corresponding data.

Some of the CAN Data Frame fields have been left out of the PDU definition because they are controlled entirely by the CAN specification and are invisible to all of the OSI layers above the data link layer. They include the SOF, SRR, IDE, RTR, parts of the control field, CRC field, ACK field, and the EOF field. These fields are defined by the CAN protocol definition and are not modified by SAE J1939. The seven PDU fields are illustrated in Figure 3. Each of the fields within the PDU is defined in the subsequent sections.



Definitions: P is Priority, EDP is Extended Data Page, DP is Data Page, PF is PDU Format, PS is PDU Specific, and SA is Source Address

FIGURE 3 - PROTOCOL DATA UNIT

5.2.1 Priority (P)

These three bits are used to optimize message latency for transmission onto the bus only. They should be globally masked off by the receiver (ignored). The priority of any message can be set from highest, 0 (000_2), to lowest, 7 (111_2). The default for all control oriented messages is 3 (011_2). The default for all other informational, proprietary, request, and ACK messages is 6 (110_2). This permits the priority to be raised or lowered in the future as new PGNs are assigned and bus traffic changes. A recommended priority is assigned to each PGN when it is added to the application layer document. However, the priority field should be reprogrammable to allow for network tuning by the OEM should the need arise.

5.2.2 Extended Data Page (EDP)

The Extended Data Page bit is used in conjunction with the Data Page bit to determine the structure of the CAN identifier of the CAN data frame. All J1939 messages should set the Extended Data Page bit to ZERO on transmit. See Table 3, Definition of Extended Data Page and Data Page Use, to find the defined uses of the EDP and DP fields. Future definitions might possibly be expanding the PDU Format field, defining new PDU formats, or increasing the address space.

5.2.3 Data Page (DP)

The Data Page Bit is used in conjunction with the Extended Data Page bit to determine the structure of the CAN identifier of the CAN data frame. With the EDP set to 0 the data page bit selects between page 0 and page 1 of Parameter Group Number (PGN) descriptions. See Table 3. Also see Figure 4 to see the SAE J1939 Parameter Group Number template.

TABLE 3 - DEFINITION OF EXTENDED DATA PAGE AND DATA PAGE USE

Extended Data Page bit 25 CAN ID bit 25	Data Page bit 24 CAN ID bit 24	Description
0	0	SAE J1939 page 0 PGNs
0	1	SAE J1939 page 1 PGNs
1	0	SAE J1939 reserved
1	1	ISO 15765-3 defined

Table 3 Note: The EDP (Extended Data Page) and DP (Data Page) of the CAN 29 bit identifier being set to “11” identifies it as an ISO 15765-3 message. This means that the rest of the CAN identifier is NOT set up the same as J1939 specifies. CAN frames following this format are not described in the J1939 series of SAE recommended practices.

Legend:

EDP	= Extended Data Page	(1 bit)			
DP	= Data Page	(1 bit)	GE	= Group Extension	(8 bits)
PF	= PDU Format	(8 bits)	P	= Priority	(1 bit)
PS	= PDU Specific Field	(8 bits)	NA	= Not Allowed	
DA	= Destination Address	(8 bits)	Un	= Undefined	
PGN	= Parameter Group Number	(3 bytes)			

EDP	DP	PF	PS	Parameter Group Definition	Multipacket	PGN
0	0	0	DA	PDU1 Format - 100 mS or less	NA	000
0	0	1	DA			256
	.					
	Boundary x					
0	0	238	DA	PDU1 Format - Greater than 100 mS	Allowed	60928
0	0	239	DA	PDU1 Format - Proprietary A	Allowed	61184
0	0	240	0	PDU2 Format - 100 mS or less	NA	61440
0	0	240	1			61441
	.					
	Boundary y					
	.					
0	0	254	254			65278
0	0	254	255	PDU2 Format - Greater than 100 mS	Allowed	65279
0	0	255	un	PDU2 Format - Proprietary B	Allowed	65280-65535
0	1	0	DA	PDU1 format - 100 mS or less	NA	65536
0	1	1	DA			65792
	.					
	Boundary x1					
	.					
0	1	238	DA	PDU1 Format - Greater than 100 mS	Allowed	126464
0	1	239	DA	PDU1 Format - Proprietary A2	Allowed	126720
0	1	240	0	PDU2 format - 100 mS or less	NA	126976
0	1	240	1			126977
	.					
	Boundary y1					
	.					
0	1	255	253			
0	1	255	254	PDU2 Format - Greater than 100 mS	Allowed	131070
0	1	255	255	PDU2 Format - Greater than 100 mS	Allowed	131071

FIGURE 4 - SAE J1939 PARAMETER GROUP NUMBER TEMPLATE

5.2.4 PDU Format (PF)

The PDU Format is an 8-bit field that determines the PDU format and is one of the fields used to determine the Parameter Group Number assigned to the data field. Parameter Group Numbers are used to identify or label commands, data, some requests, acknowledgments, and negative-acknowledgments. Parameter Group Numbers identify or label information that may require one or more CAN Data Frames to communicate the information. If there is more information than fits in 8 data bytes, then a multipacket message needs to be sent (see Section 5.10). If there are 8 or less data bytes, then a single CAN data Frame is used. A Parameter Group Number can represent one or more parameters, where a parameter is a piece of data such as engine rpm. Even though a Parameter Group Number label can be used for one parameter, it is recommended that multiple parameters be grouped so that all 8 bytes of the data field are used.

The definition of proprietary Parameter Group Numbers has been established allowing both PDU1 and PDU2 formats to be used. The interpretation of the proprietary information varies by manufacturer. For example even though two different engines may use the same source address, manufacturer “A’s” proprietary communications is more likely to be different from manufacturer “B’s.”

5.2.5 PDU Specific (PS)

This is an 8-bit field and its definition depends on the PDU format. Depending on the PDU format it can be a Destination Address or a Group Extension. If the value of the PDU Format field is below 240, then the PDU specific field is a destination address. If the value of the PF field is 240 to 255, then the PDU specific field contains a Group Extension value. See Table 4. Refer to Figure 4 for the range of Parameter Group Numbers.

TABLE 4 - PDU SPECIFIC

	PDU Format Field	PDU Specific Field
PDU1 Format	0-239	Destination Address
PDU2 Format	240-255	Group Extension

5.2.5.1 Destination Address (DA)

This field defines the specific address to which the message is being sent. Note that any other device should ignore this message. The global destination address (255) requires all devices to listen and respond accordingly as message recipients.

5.2.5.2 Group Extension (GE)

The Group Extension field, in conjunction with the four least significant bits of the PDU Format field (note that when the four most significant bits of the PDU Format field are set, it indicates that the PS field is a Group Extension), provides for 4096 Parameter Groups per data page. These 4096 Parameter Groups are only available using the PDU2 format. In addition, 240 Parameter Groups are provided in each data page for use only in the PDU1 format. In total, 8672 Parameter Groups are available to be defined using the two data pages currently available.

The total number of Parameter Groups available can be calculated as follows in Equation 1:

$$(240 + (16 \times 256)) \times 2 = 8672 \quad \text{(Eq.1)}$$

where:

240 = Number of PDU Format field values available per data page (i.e. PDU1 Format, PS Field equals Destination Address)
16 = PDU Format values per Group Extension value (i.e. PDU2 Format)
256 = Number of possible Group Extension values (i.e. PDU2 Format)
2 = Number of Data Page states (both PDU Formats)

5.2.6 Source Address (SA)

The Source Address field is 8 bits long. There shall only be one device on the network with a given source address. Therefore, the source address field assures that the CAN identifier is unique, as required by CAN. Address management and allocation is detailed in SAE J1939-81. Procedures are defined in SAE J1939-81 to prevent duplication of source addresses. Reference SAE J1939 Appendix B, Tables B2 through B9, for source address assignments.

5.2.7 Data Field

5.2.7.1 Data from 0 to 8 Bytes

When 8 bytes or less of data are required for expressing a given Parameter Group, then all 8 data bytes of the CAN data frame can be used. It is generally recommended that 8 data bytes be allocated or reserved for all Parameter Group Number assignments which are likely to expand in the future. This provides a means to easily add parameters and not be incompatible with previous revisions that only defined part of the data field. Once the number of data bytes associated with a Parameter Group Number is specified, the number of data bytes cannot be changed (cannot become multipacket either unless originally defined as multipacket). The CAN Data Length Code (DLC) is set to the defined Parameter Group "Data Length" value when it is 8 bytes or less; otherwise when the PG Data Length is 9 or greater, the CAN DLC is set to 8. For example the REQUEST PGN, 59904, has the PG Data Length as 3 so the CAN DLC is set to 3. It is important to note that an individual group function Parameter Group (see Section 5.4.5) must use the same length data field because the CAN identifier will be identical while the CAN data field will be used to convey the specific group subfunctions, therefore requiring many different interpretations based on the CAN data field.

5.2.7.2 Data from 9 Up to 1785 Bytes

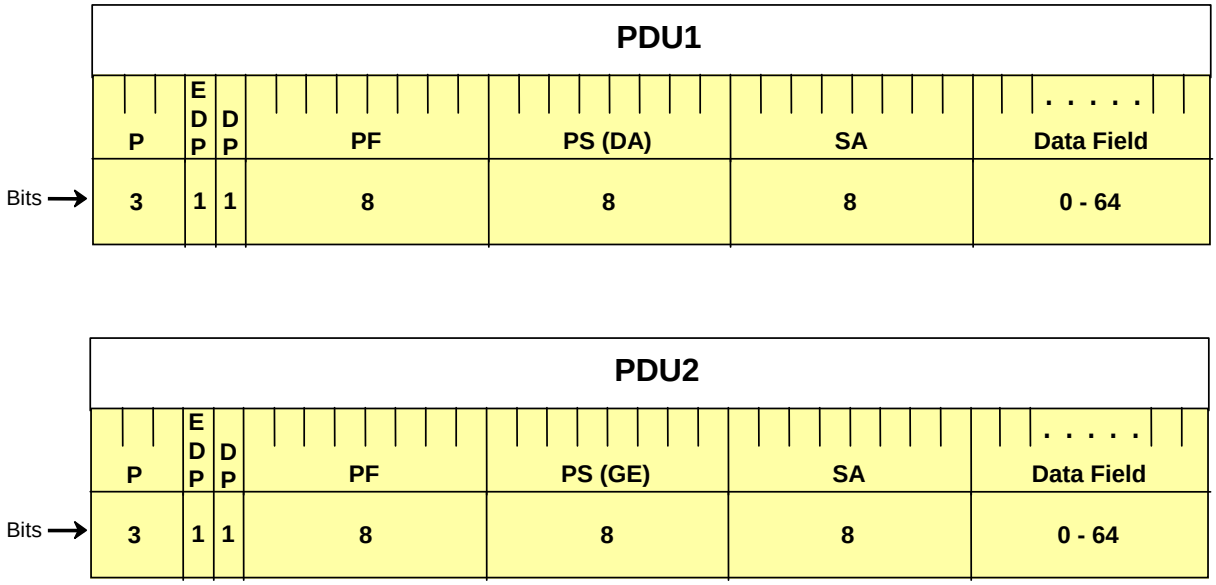
When 9 up to 1785 data bytes are needed to express a given Parameter Group, the communication of this data is done in multiple CAN Data Frames. Thus, the term multipacket is used to describe this type of Parameter Group Number. A Parameter Group defined as multipacket capable, having fewer than 9 data bytes to transfer in a specific instance, shall be sent in a single CAN Data Frame with the DLC set to 8. When a particular Parameter Group has 9 or more data bytes to transfer, the "Transport Protocol Function" is used. The Transport Protocol Function's Connection Management capability is used to set up and close out the communication of the multipacket Parameter Groups. The Transport Protocol Data Transfer capability is used to communicate the data itself in a series of CAN Data Frames (packets) containing the packetized data. Additionally, the Transport Protocol Function provides flow control and handshaking capabilities for destination specific transfers. See Section 5.10.

All CAN Data Frames associated with a particular multipacket response are required to have a DLC of 8. All unused data bytes are set to "not available" (see SAE J1939-71). The number of bytes per packet is fixed; however, SAE J1939 defines multipacket messages that have a variable and/or a fixed number of packets. The Parameter Group Number for active diagnostic codes is an example of a multipacket message that has a "variable" number of packets. Parameter Groups that are defined as multipacket only use the transport protocol when the number of data bytes to send exceeds eight.

5.3 Protocol Data Unit (PDU) Formats

The available PDU formats are illustrated in Figure 5. Two PDU formats are defined: PDU1 Format (PS = Destination Address) and PDU2 Format (PS = Group Extension). PDU1 Format allows for direction of the CAN Data Frame to a specific destination address (device). The PDU2 Format can only communicate CAN Data Frames that are not destination specific. The creation of two separate PDU formats was established in order to provide more possible Parameter Group Number combinations while still providing for destination specific communications. Proprietary Parameter Group definitions have been assigned so that both PDU formats are available for proprietary communications. A standardized method was established for proprietary communications to prevent possible conflicts in identifier usage.

The definition of proprietary Parameter Group Numbers has been established allowing both PDU1 and PDU2 Formats to be used. The interpretation of the proprietary information varies by manufacturer. For example engine manufacturer "A's" proprietary communications are likely to be different than engine manufacturer "B's" even though they both use the same source address.



Definitions: P is Priority, EDP is Extended Data Page, DP is Data Page, PF is PDU Format, PS is PDU Specific, DA is Destination Address, GE is Group Extension, and SA is Source Address

FIGURE 5 - AVAILABLE PDU FORMATS

5.3.1 PDU1 Format

This format allows applicable Parameter Groups to be sent to either a specific or global destination. The PDU Specific field contains a destination address. PDU1 Format messages can be requested or sent as unsolicited messages.

PDU1 Format messages are determined by the PDU Format field. When the PDU Format field value is 0 to 239, the message is a PDU1 Format. The format of the PDU1 message is illustrated in Figure 5. Also see Figure 6, PDU1 Format.

Parameter Groups, requiring a destination (PDU1) and minimal latency, start at PF = 0 and increment toward x (or x1). Reference Figure 4.

Parameter Groups, requiring a destination where latency is not critical, start at PF = 238 and decrement toward x (or x1). Reference Figure 4.

A PF equal to 239, extended data page bit = 0, and data page bit = 0 or 1, is assigned for proprietary use. In this case the PDU Specific field is a destination address (see Section 5.4.5). The Proprietary A PGN is 61184 and the Proprietary A2 PGN is 126720.

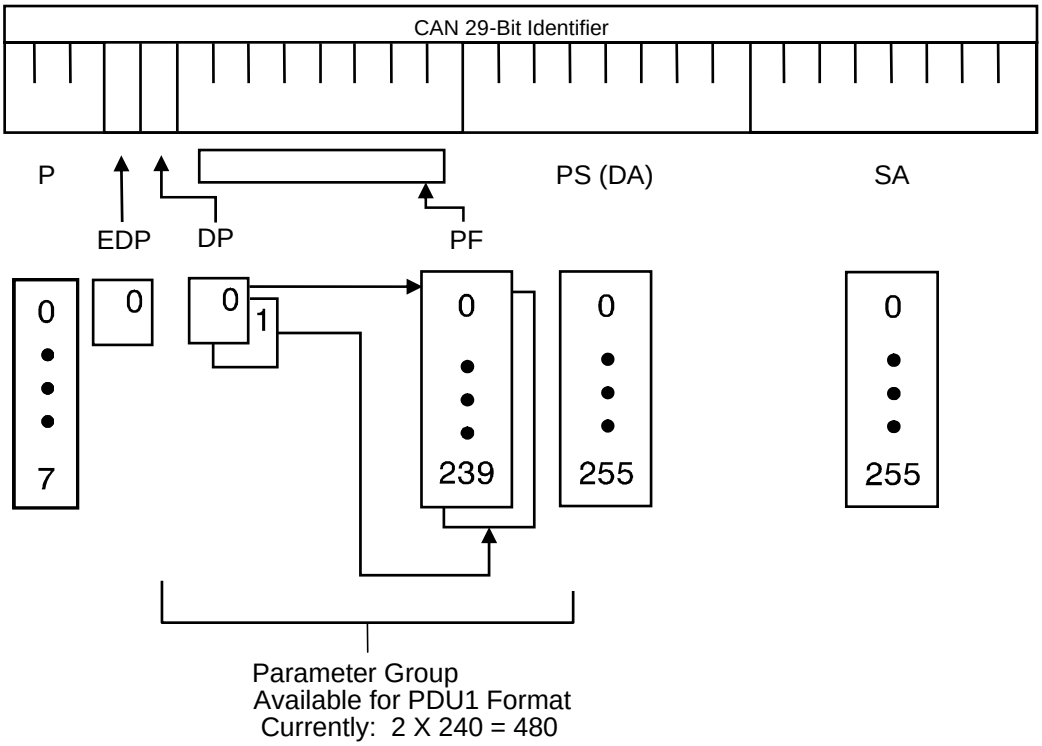


FIGURE 6 - PDU1 FORMAT

5.3.2 PDU2 Format

This format can only be used to communicate Parameter Groups as global messages. PDU2 Format messages can be requested or sent as unsolicited messages. Selection of PDU2 Format, at the time a PGN is assigned, prevents that PGN from ever being able to be directed to a specific destination. The PDU Specific contains a Group Extension.

PDU2 Format messages are defined to be those where the PDU Format value is equal to 240 to 255 (see Figure 4). The format of the PDU2 message is illustrated in Figure 5. Also see Figure 7, PDU2 Format.

The Parameter Group Number of messages that are sent at fast update rates (generally less than 100 ms) start at PF = 240 and increment toward y (or y1). Reference Figure 4.

The Parameter Group Number of messages that are only requested, sent on change, or sent at slow update rates (generally greater than every 100 ms) start at PF = 254 and decrement toward y (or y1). Reference Figure 4.

A PF equal to 255 (Extended Data Page = 0 and Data Page bit = 0) is assigned for proprietary use. In this case the PDU Specific field is left to be defined and used by each manufacturer (see Section 5.4.5). The PGN for Proprietary B covers the range 65280 to 65535.

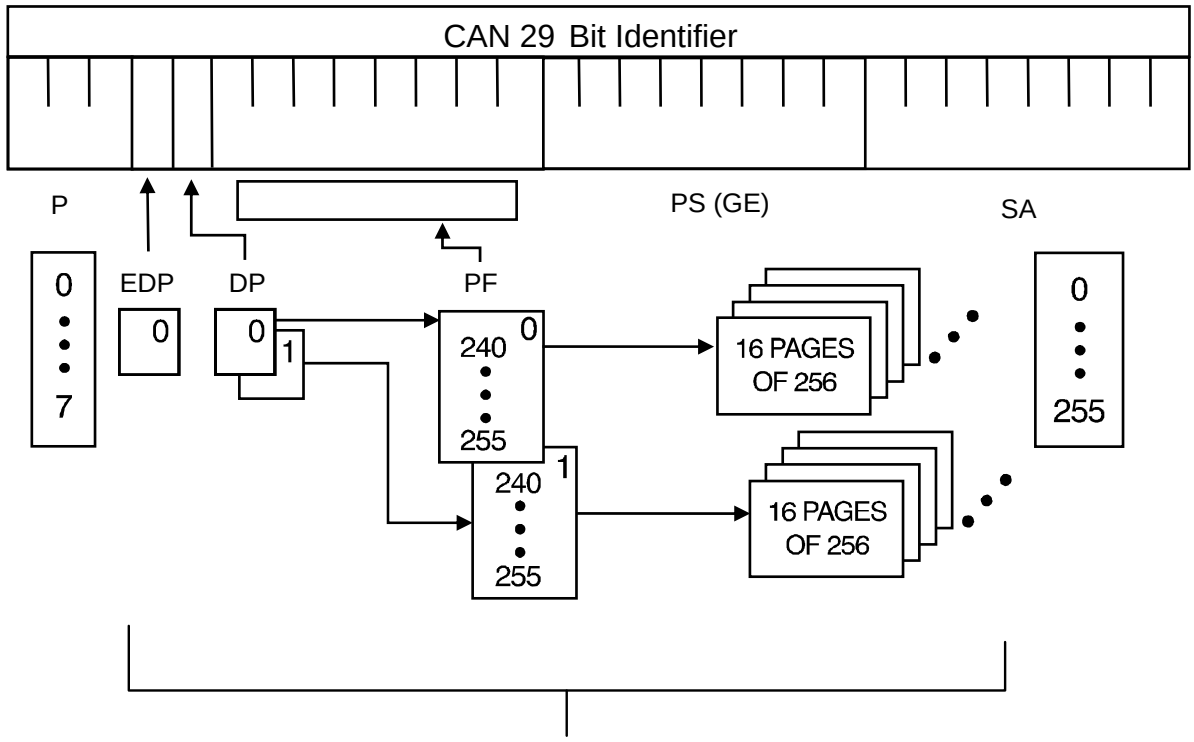


FIGURE 7 - PDU2 FORMAT

5.4 Message Types

There are five message types currently supported. These types are the following: Commands, Requests, Broadcasts/Responses, Acknowledgment, and Group Functions. The specific message type is recognized by its assigned Parameter Group Number. Refer to Appendix A of SAE J1939 for examples of PGN assignments. The RTR bit (defined in the CAN protocol for remote frames) is not to be used in the recessive state (logical 1). Therefore, Remote Transmission Requests (RTR = 1) are not available for use in SAE J1939.

For multibyte parameters that appear in the data field of a CAN Data Frame they shall be placed least significant byte first. Exceptions are noted where applicable (i.e. ASCII data). So if a 2-byte parameter were to be placed in bytes 7 and 8 of the CAN DATA Frame, the LSB would be placed in byte 7 and the most significant byte in byte 8.

5.4.1 Command

This message type categorizes those Parameter Groups that convey a command to a specific or global destination from a source. The destination is then supposed to take specific actions based on the reception of this command message type. Both PDU1 (PS = Destination Address) and PDU2 Format (PS = Group Extension) messages can be used for commands. Example command type messages may include “Transmission Control,” “Address Request,” “Torque/Speed Control,” etc.

5.4.2 Request

This message type, identified by the PGN, provides the capability to request information globally or from a specific destination. Requests specific to one destination are known as destination specific requests. The information in Figure 8 shows the PGN definition for the “Request PGN” Parameter Group. The information is in the same format as Parameter Group definitions in SAE J1939-71. Table D1 in Appendix D has the Suspect Parameter Number (SPN) assignments to parameters in this message.

Parameter Group Name:	**Request**
Definition:	Used to request a Parameter Group from a network device or devices.
Transmission repetition rate:	Per user requirements, generally recommended that requests occur no more than 2 or 3 times per second.
Data length:	3 bytes (The CAN frame for this PG shall set the DLC to 3.)
Extended Data Page:	0
Data page:	0
PDU Format:	234
PDU specific field:	Destination Address (global or specific)
Default priority:	6
Parameter Group Number:	59904 ($00EA00_{16}$)
Byte: 1,2,3	Parameter Group Number being requested (see Section 5.1.2 for field definition and byte order)

FIGURE 8 - REQUEST PGN DEFINITION

Table 5 iterates the request/response possibilities for PDU1 and PDU2 Format PGNs. It clarifies that the transmitter of a message determines whether the destination is specific or global based on whether the request was to a specific destination address or to a global destination address.

Table 5 also shows that for unsolicited messages the transmitter has the choice of transmitting to a specific destination address or to a global destination address for PDU1 PGNs and PDU2 PGNs with more than 8 bytes. For PDU2 PGNs with 8 bytes or less, the transmitter can only send the data globally. See Table 5.

TABLE 5 - PDU1 AND PDU2 TRANSMIT, REQUEST AND RESPONSE REQUIREMENTS

PDU Format of Requested PGN	Data Length of Requested PGN	Request PGN 59904	Response	TP Used
1	≤ 8 bytes	DA Specific	DA Specific	NA
1	≤ 8 bytes	DA Global	DA Global	NA
1	≤ 8 bytes	none	DA Global	NA
			DA Specific	NA
1	> 8 bytes	DA Specific	DA Specific	RTS/CTS
1	> 8 bytes	DA Global	DA Global	BAM
1	> 8 bytes	none	DA Global	BAM
			DA Specific	RTS/CTS
2	≤ 8 bytes	DA Specific	DA Global	NA
2	≤ 8 bytes	DA Global	DA Global	NA
2	≤ 8 bytes	none	DA Global	NA
2	> 8 bytes	DA Specific	DA Specific	RTS/CTS
2	> 8 bytes	DA Global	DA Global	BAM
2	> 8 bytes	none	DA Global	BAM
			DA Specific	RTS/CTS

Notes to Table 5 — General rules of operation for determining whether to send a PGN to a global or specific destination:

1. If the Request is sent to a global address, then the response is sent to a global address.
   a) NOTE: A NACK is not permitted as a response to a global request.
2. If the Request is sent to a specific address, then the response is sent to a specific address.
   a) NOTE: A NACK is required if the PGN is not supported.
   b) If the data length is more than 8 bytes, the Transport Protocol RTS/CTS must be used for the response to a specific address.
   c) Exceptions:
      i) PDU2 format PGNs with 8 bytes or less can only be sent to a global destination because there is no destination address field in the PDU2 Format.
      ii) The Address Claim PGN is sent to the global destination address even though the request for it may have been to a specific destination address (see J1939-81).
      iii) The Acknowledgment PGN response uses a global destination address even though the PGN that causes Acknowledgment was sent to a specific destination address.
3. For periodic broadcasts or unsolicited messages
   a) PDU1 Format PGNs or PDU2 Format PGNs can be sent to a global or specific destination address.
      i) Exception: PDU2 Format PGNs with 8 bytes or less can only be sent to a global destination because there is no destination address field in the PDU2 Format.
4. Exceptions to these rules do exist and are specifically noted when this is the case. These exceptions are noted in the applicable document and section where the PGN is defined. Two different types of exceptions are included:
   a) When the response destination address does not specify the source address of the request. Some examples have been noted above (e.g. Address Claim PGN and Acknowledgment PGN).
   b) When the PGN does not support all forms of the available addressing. That is, some PGNs may not be designed to support the address capability that is available for PDU1 or PDU2 Format messages.

Table 6 contains two examples showing how the Request PGN is used.

TABLE 6 - USE OF THE SPECIFIED FIELDS IN SAE J1939 PDU1 FORMAT

Message Type	PGN	PS (DA)	SA	Data 1	Data 2	Data 3
Global Request	59904	255 Responders	SA1 Requester	PGN LSB[1]	PGN	PGN MSB[1]
Specific Request	59904	SA2 Responder	SA1 Requester	PGN LSB[1]	PGN	PGN MSB[1]

[1] The Parameter Group Number in the data field is used to identify the information being requested.

A response is always required from the specified destination (not global) even if it is a NACK indicating that the particular PGN value is not supported. Note that some PGNs are multipacket, so several CAN Data Frames can occur as a result of a single request.

A global request shall not be responded to with a NACK when a particular PGN is not supported by a node.

The Request PGN can be directed to a specific destination address to find out if a specific Parameter Group is supported (i.e. does the requested destination address transmit the specific PG?) The response to the request determines whether the PGN is supported. If it is supported, then the responding device shall send the requested information. If the Acknowledgment PGN is appropriate, then the control byte shall be set to 0 or 2 or 3. If it is not supported, the responding device sends the Acknowledgment PGN with the control byte set to one, for Negative Acknowledgment. The remaining portions of the SAE J1939 PDU format and Parameter Group must be filled in appropriately (reference Section 5.4.4). Note that per the definitions in this paragraph the phrase “not supported” means that the PG is not transmitted. It is not possible to determine whether a device will act upon (when received) the PG by using this method.

If a device fails to get a response (either the PGN or NACK) to a Request within the Response timeout, then the device may resend or retry the same Request. The number of retries for a specific Request should be limited to two (2) retries, i.e. the Request is issued a total of three (3) times. If the device fails to get a response (either the PGN or NACK) to the Request after the third retry, then the device should abandon further request attempts for the same information or the device may wait for an extended period of time (minutes rather than seconds) before attempting to request the same information.

5.4.3 Broadcast/Response

This Message Type can be an unsolicited broadcast of information from a device or it can be the response to a Command or a Request.

5.4.4 Acknowledgment

There are two forms of acknowledgment available. The first form is provided for by the CAN protocol. It consists of an "in-frame" acknowledgment which confirms that a message has been received by at least one node. In addition messages are further acknowledged by the absence of CAN error frames. Their absence acknowledges that all other powered and connected devices received the message correctly.

The second form of acknowledgment is the response of a "normal broadcast" or "ACK" or "NACK" to a specific command or request as provided for by an application layer. The definition of the Acknowledgment Parameter Group is contained in Figure 9. Table D1 in Appendix D has the Suspect Parameter Number assignments to parameters in this message. The type of acknowledgment required for some Parameter Groups is defined in the applications layer.

For Group Function PGs (see Section 5.4.5) the Group Function Value parameter allows a device to identify the specific group function that is being acknowledged. The Group Function Values are unique to each Group Function PG. It is desirable that the Group Function Values only use the range 0 to 250.

Each form of Acknowledgment includes an address acknowledge byte that contains the source address of the originator of the request that the Acknowledgment is directed towards. Since the Acknowledgment PGN is always directed to the Global Address, these parameters allow the receiver to know the target being acknowledged. In Figure 9 these parameters, in byte 5, are Address Acknowledged, Address Negative Acknowledgment, Address Access Denied, and Address Busy.

Parameter Group Name:		**Acknowledgment**
Definition:		The Acknowledgment PG is used to provide a handshake mechanism between transmitting and receiving devices.
Transmission repetition rate:		Upon reception of a Parameter Group Number that requires this form of acknowledgment.
Data length:		8 bytes
Extended Data Page:		0
Data Page:		0
PDU Format:		232
PDU Specific:		Destination address[1] = Global (255)
Default priority:		6
Parameter Group Number:		59392 ($00E800_{16}$)
Data ranges for parameters used by this Message Type:		
Control byte:	0 to 3	See definitions below
	4 to 255	Reserved for assignment by SAE
Group Function Value	0-250	Definition is specific to the individual PGN, when applicable. Most often it is located as the first byte in the data field of the applicable Group Function PG.
	251 to 255	Follows conventions defined in J1939-71
Positive Acknowledgment: Control byte = 0		
Byte:	1	Control byte = 0, Positive Acknowledgment (ACK)
	2	Group Function Value (If applicable)
	3-4	Reserved for assignment by SAE, these bytes should be filled with FF_{16}
	5	Address Acknowledged
	6	Parameter Group Number of requested information (8 LSB of parameter group number, bit 8 most significant)
	7	Parameter Group Number of requested information (2nd byte of parameter group number, bit 8 most significant)
	8	Parameter Group Number of requested information (8 MSBs of parameter group number, bit 8 most significant)
Negative Acknowledgment: Control byte = 1		
Byte:	1	Control byte = 1, Negative Acknowledgment (NACK)
	2	Group Function Value (if applicable)
	3-4	Reserved for assignment by SAE, these bytes should be filled with FF_{16}
	5	Address Negative Acknowledgement
	6-8	Parameter Group Number of requested information (see above)
Access Denied: Control byte = 2		
Byte:	1	Control byte = 2, Access Denied (PGN supported but security denied access)
	2	Group Function Value (if applicable)
	3-4	Reserved for assignment by SAE, these bytes should be filled with FF_{16}
	5	Address Access Denied
	6-8	Parameter Group Number of requested information (see above)
Cannot Respond: Control byte = 3		
Byte	1	Control byte = 3, Cannot Respond (PGN supported but ECU is busy and cannot respond now. Re-request the data at a later time.)
	2	Group Function Value (if applicable)
	3-4	Reserved for assignment by SAE, these bytes should be filled with FF_{16}
	5	Address Busy
	6-8	Parameter Group Number of requested information (see above)

FIGURE 9 - ACKNOWLEDGMENT PGN DEFINITION

5.4.5 Group Function

This Message Type is used for groups of special functions (e.g. proprietary functions, network management functions, multipacket transport functions, etc.). Each group function is recognized by its assigned PGN. See Figure 10, Figure 11

[1] The global destination address makes it possible to filter on one CAN Identifier for all Acknowledgment messages.

and Figure 12. Table D1 in Appendix D has the Suspect Parameter Number assignments to parameters in these messages. The function itself is defined within the data structure (typically the first byte of the data field). More detailed explanation of the group functions proprietary and transport protocol will be given in the sections to follow. The proprietary group function provides a means to transmit proprietary messages in a way that eliminates CAN Identifier usage conflicts between different manufacturers. It also provides a means for receiving and distinguishing proprietary messages for use when desired. Group Functions may need to provide their own request, ACK, and/or NACK mechanisms if the messages defined in J1939-21 are not sufficient.

A request using PGN 59904 (reference Section 5.4.2) can be used to find out if a specific Parameter Group of the message type, Group Function, is supported. If it is supported, then the responding device sends the Acknowledgment PGN with the control byte equal to zero, for Positive Acknowledgment, or equal to two, Access Denied or equal to three, Cannot Respond. If it is not supported, the responding device sends the Acknowledgment PGN with the control byte set to one, for Negative Acknowledgment. The remaining portions of the SAE J1939 PDU format and Parameter Group must be filled in appropriately (reference Section 5.4.4). Note that per the definitions in this paragraph the phrase “not supported” means that the PG is not transmitted. It is not possible to determine whether a device will act upon (when received) the PG by using this method.

Parameter Group Name:	**Proprietary A**
Definition:	This proprietary PG uses the destination specific PDU format allowing manufacturers to direct their proprietary communications to a specific destination node. How the data field of this message is used is up to each manufacturer. Use of proprietary messages is at the manufacturer’s discretion with the constraint that significant percentages (2 percent or more) of vehicle network utilization must be avoided.
Transmission repetition rate:	Per user requirements
Data length:	0 to 1785 bytes (multipacket supported)
Extended Data Page:	0
Data Page:	0
PDU Format:	239
PDU Specific:	Destination Address
Default priority:	6
Parameter Group Number:	61184 ($00EF00_{16}$)
Byte: 1-8	Manufacturer specific use

Data ranges for parameters used by this Group Function:
None defined by SAE

FIGURE 10 - PROPRIETARY A PGN DEFINITION

Parameter Group Name:	**Proprietary A2**
Definition:	This proprietary PG uses the destination specific PDU format allowing manufacturers to direct their proprietary communications to a specific destination node. How the data field of this message is used is up to each manufacturer. Use of proprietary messages is at the manufacturer's discretion with the constraint that significant percentages (2 percent or more) of vehicle network utilization must be avoided.
Transmission repetition rate:	Per user requirements
Data length:	0 to 1785 bytes (multipacket supported)
Extended Data Page:	0
Data Page:	1
PDU Format:	239
PDU Specific:	Destination Address
Default priority:	6
Parameter Group Number:	126720 ($01EF00_{16}$)
Byte: 1-8	Manufacturer specific use

Data ranges for parameters used by this Group Function:
None defined by SAE

FIGURE 11 - PROPRIETARY A2 PGN DEFINITION

Parameter Group Name:	**Proprietary B**
Definition:	This proprietary PG uses the PDU2 Format message allowing manufacturers to define the PS (GE) field content as they desire. However, significant percentages (2 percent or more) of vehicle network utilization must be avoided. How the PS (GE) and data fields of this message are used is up to each manufacturer. The data length of these messages has been left up to each manufacturer. Therefore, two manufacturers, say of transmissions, may use the same GE value and it may very well have a different Data Length Code. Receivers of this information would need to differentiate between the two manufacturers.
Transmission repetition rate:	Per user requirements
Data length:	0 to 1785 bytes (multipacket supported)
Extended Data Page:	0
Data Page:	0
PDU Format:	255
PDU Specific:	Group Extension (manufacturer assigned)
Default priority:	6
Parameter Group Number:	65280 to 65535 ($00FF00_{16}$ to $00FFFF_{16}$)
Byte: 1-8	Manufacturer defined usage

Data ranges for parameters used by this Group Function:
Manufacturer defined usage allows the Data Length Code being different per component supplier and source address. Caution should be used when using the Proprietary B Parameter Group (PGN = 65280) because multiple source addresses can use the same Proprietary B PGN value for different purposes.

FIGURE 12 - PROPRIETARY B PGN DEFINITION

5.4.6 Request2

The Request2 PG provides the capability for the requestor to specify whether the responder should use the Transfer PGN 51712. By identifying that the responder should use the Transfer PGN, it provides the ability for the responder to report the data set for all devices it is tasked with reporting (see Section 0). All devices include what the responding device would normally report upon receiving the same PGN requested in PGN 59904 (properly formatted for TRANSFER PGN) and the data set for each device it is tasked to report. For instance if the "Use Transfer PGN" parameter is yes (01), the

response shall include all data known relative to the requested PGN. When "Use Transfer PGN" is 00, the effect of the Request2 PGN is the same as the Request PGN (59904). The response to the Request2 when the "Use Transfer PGN" equals 00 is not sent using the "Transfer" PGN and it is sent exactly the same as it would be had the request been made using the "Request" PGN (i.e. PGN 59904). See Figure 13 below. Table D1 in Appendix D has the SPN assignments to parameters in this message.

The "Request2" and "Transfer" PGNs are required in cases where a given device is tasked with reporting a PGN and data about more than one device. Examples include PGNs such as Vehicle Identification, Component ID and Software Identification.

If a device fails to get a response (either the PGN or NACK) to a Request within the Response timeout, then the device may resend or retry the same Request. The number of retries for a specific Request should be limited to two (2) retries, i.e. the Request is issued a total of three (3) times. If the device fails to get a response (either the PGN or NACK) to the Request after the third retry, then the device should abandon further request attempts for the same information or the device may wait for an extended period of time (minutes rather than seconds) before attempting to request the same information.

The support of REQUEST2 is an option.

Parameter Group Name:	**Request2**
Definition:	Used to Request a PGN from network device or devices and to specify whether the response should use the Transfer PGN or not.
Transmission repetition rate:	Per user requirements, generally recommended that requests occur no more than 2 or 3 times per second. When a device supports Request2, a NACK is required if the PG being asked for with a destination specific address is not supported (see PGN 59392).
Data length:	8 bytes (multipacket supported)
Extended Data Page:	0
Data Page:	0
PDU Format:	201
PDU Specific:	Destination Address (global or specific)
Default priority:	6
Parameter Group Number:	51456 ($00C900_{16}$)
Bytes 1-3:	Requested PGN
Byte 4:	Special Instructions
Bits 8-3:	Reserved for SAE assignment
Bits 2-1:	Use Transfer PGN for response (00 = No, 01=Yes, 10 = undefined, 11 = NA)
Bytes 5-8:	Reserved for SAE assignment

FIGURE 13 - REQUEST2 PGN FORMAT

5.4.7 Transfer

The Transfer PGN provides a mechanism for reporting multiple data sets for a given PGN in response to a Request2 (see Section 5.4.6). See Figure 14 for the definition of a "data set." These multiple sets of data for a given PGN require that each data set have a length and be labeled with four bytes from the J1939-81 Name. The four bytes of the Name identify each device. The device responding to the request shall report the same information it would with PGN 59904 as the first data set in this response. If a device only has one data set, then it shall respond with the one data set utilizing the Transfer PG.

The "Request2" and "Transfer" PGNs are useful in cases where a given device is tasked with reporting a PGN and data about more than one device. Examples include PGNs such as Vehicle Identification, Component ID and Software Identification.

See Figure 14 below. Table D1 in Appendix D has the SPN assignments to parameters in this message.

Parameter Group Name:		**Transfer**
Definition		Used for transfer of data in response to a Request2 when the "Use Transfer PGN for response" is set to "Yes"
Transmission repetition rate:		In response to a Request2 PGN with "Use Transfer PGN" = 01
Data length:		9 to 1785 bytes (multipacket supported)
Extended Data Page:		0
Data Page:		0
PDU Format:		202
PDU Specific:		Destination Address (global or specific)
Default priority:		6
Parameter Group Number:		51712 ($00CA00_{16}$)
Bytes 1-3:		(a) PGN requested by Request2 (See Table 2 for PGN ordering)
Byte 4:		(b) Length of data for the reported PGN associated to the device identified (e.g. Controller Application Identity in bytes 5-8). Length value is the total of this byte, length of identity bytes (i.e. bytes 5-8), and the associated PGN data. So the length is b + c + d.
Bytes 5-8:		(c) Identity of device associated to the PGN and data (i.e. Controller Application Identity) SAE J1939-81 defines the four bytes of the Name used here in bytes 5 through 8.
Byte 5:	Bits 8-4	Function Instance (most significant at bit 8)
	Bits 3-1	ECU Instance (most significant at bit 3)
Byte 6:	Bits 8-1	Function (most significant at bit 8)
Byte 7:	Bits 8-2	Vehicle System (most significant at bit 8)
	Bit 1	Reserved
Byte 8:	Bit 8	Arbitrary Address Capable
	Bits 7-5	Industry Group (most significant at bit 7)
	Bits 4-1	Vehicle System Instance (most significant at bit 4)
Bytes 9-x:		(d) Data of reported PGN in bytes 1-3
Bytes x+1 to n		Repeating information for 2nd and following shall contain: "Controller Application Identity," Length, and "PGN requested by Request2's data." See format definition below.

Format:

a,b,c,d,b,c,d,b,c,d...

a: PGN requested by Request2 when "transfer mode" is set to yes
b: First Data Set: Length of concatenated Controller Application Identity and associated PGN data
The length = b + c + d
c: Identity of Controller Applicaton to which field "d" is associated
d: Requested PGN's data for specific Controller Application
b: Second Data Set: Length of concatenated Controller Application Identity and associated PGN data
c: Identity of Controller Application to which field "d" is associated
d: Requested PGN's data for second specific Controller Application
... etc.

Example: For a given vehicle the engine ECU knows the VIN numbers for the tractor and the trailer. Another device sends the Request2 directed to the global destination, requesting the VIN with "Use Transfer PGN" set to 01. The response from the engine might be:

BAM transfer of the "Transfer" PGN reporting the VIN for the tractor and VIN for the trailer

If the request had the "Use Transfer PGN" set to 00, the response would be:

BAM transfer of the VIN for the tractor but not utilizing the Transfer PGN

FIGURE 14 - TRANSFER PGN FORMAT

5.5 Message Priority

The CAN data frame priority shall be per CAN specification 2.0 Part B. The value within the CAN identifier field determines the message priority. A low value (the 29-bit identifier being all zeros) has a high priority, while the largest CAN identifier has the lowest priority (the 29-bit identifier being all ones). The assignments are identified in the

applications layer following the guidelines spelled out in Section 5.9 of this document and as defined in SAE J1939 top level document (SAE J1939).

5.6 Bus Access

When the bus is free, any node may start to transmit a frame. If two or more nodes start to transmit frames at the same time, the bus access conflict is resolved by contention-based arbitration using the CAN Data Frame Identifier. The mechanism of arbitration guarantees that neither information nor time is lost. The transmitter with the frame of the highest priority gains bus access (reference CAN 2.0 Part B).

5.7 Contention-Based Arbitration

During arbitration, every transmitter compares the level of the bit transmitted with the level that is monitored on the bus. If these levels are equal, the node may continue to send. When a “recessive” level is sent and a “dominant” level is monitored, that node has lost arbitration and must withdraw without sending one more bit. When a “dominant” level is sent and a “recessive” level is monitored, that node detects a bit error (reference CAN 2.0 Part B).

5.8 Error Detection

The following measures are taken for detecting errors:

a. 15-bit Cyclic Redundancy Check (CRC)
b. Variable Bit Stuffing with a stuff width of 5
c. Frame format check

Transmitters compare the bit levels being transmitted with the bit level detected on the bus.

A more detailed explanation regarding these error detection techniques is available in the CAN 2.0 document identified in Section 2.1.1.

5.9 Assignment Process for Source Addresses and Parameter Group Numbers

The protocol data units that are available provide for two different formats, PDU1 and PDU2. Parameter Groups are assigned specifically to use either PDU1 or PDU2 format. Once assigned to a format, the other is not available for that Parameter Group. PDU1 format must be used anytime it is necessary to direct a Parameter Group to a specific destination. The assignment of a Parameter Group should be done while keeping in mind the following characteristics: priority, update rate, importance of the data in the packet to other network devices, and length of the data associated with the Parameter Group. In order to help with this assignment process, a request form has been created which should be used when requesting new Source Addresses or Parameter Group Number assignments (see SAE J1939 Appendix D).

Figure 4 provides a template for assigning Parameter Group Numbers. Note that the priority column is used to assign a default priority value for each PGN. The priority field should be programmable for each PGN value so that network tuning can be performed by an OEM if necessary. Although any PGN could be requested, requests are strongly discouraged for messages that are already periodically broadcast.

Messages should be assigned a Parameter Group Number that requires a destination only if it is a parameter intended to directly control (command) one of several specific devices. Otherwise a Parameter Group Number should be selected without a destination so that any device can have access to the parameters within the message.

Source Addresses are assigned in a linear fashion without concerns for message priority, update rate, or importance.

Parameter Group Numbers are assigned linearly to the various sections of Figure 4 based on the criteria provided on the Parameter Group Number and Source Address Request form. Note that multipacket messages are not permitted when the repetition rate is greater than or equal to 10 times/second.

5.9.1 Address Assignment Criteria

The number of unassigned addresses in SAE J1939 is limited and new address assignments must be made efficiently. The maximum number of addresses assigned within the system cannot exceed 256. Additions to the address definitions

therefore must be limited to significant functions within the vehicle. Examples of significant functions include the currently defined addresses for engine, transmission, brakes, fuel system, etc. Functions proposed for new address assignments within the standard should have a scope similar to currently defined addresses and should be useful to most SAE J1939 users.

Some devices use the available dynamic address allocation scheme. In this case the dynamic address can be set by the service tool and/or can be allocated at the time the network is powered up. Special precautions must be taken when using the dynamic address allocation scheme since not all network devices support this mode of operation.

5.9.2 Parameter Group Assignment Criteria

The number of unassigned Parameter Groups available in SAE J1939 is limited when compared to the large number that might be proposed for on-highway trucks and other applications. The need for large numbers of Parameter Groups is alleviated by features built into SAE J1939. Three primary communication methods exist within SAE J1939 and appropriate use of each type allows effective use of the available Parameter Group Numbers. The three communications methods are:

a. PDU1 Format (PS = Destination Address allowing destination specific communications)
b. PDU2 Format Communications (PS = Group Extension)
c. Proprietary Communications using two predefined proprietary Parameter Group Numbers

Each of the communications methods has an appropriate use. Destination specific Parameter Groups are needed where the same message must be directed to one or another destination. SAE J1939 currently defines a torque control message that may be sent to an engine. In the case of more than one engine, this message must be sent only to the desired engine and a destination specific Parameter Group is needed and has been assigned.

PDU2 Format communications apply in several situations, including:

a. Messages sent from a single or multiple sources to a single destination
b. Messages sent from a single or multiple sources to multiple destinations

PDU2 Format communications cannot be used where a message must be sent to one or another destination and not to both.

The third communications method in SAE J1939, proprietary communications, is provided by the use of the proprietary Parameter Group Numbers. A Parameter Group Number has been assigned for nondestination-specific proprietary communications and a Parameter Group Number has been assigned for destination specific proprietary communications. This allows for two functions: (a) A specific source can send its proprietary message in a PDU2 Format (nondestination specific) with the PS field identified as desired by the user; or (b) Use PDU1 Format (destination specific) allowing for situations where a service tool must direct its communication to a specific destination out of a possible group of controllers. For instance this case arises when an engine uses more than one controller but wants to be able to perform diagnostics while all of its controllers are connected to the same network. In this case the proprietary protocol needs to be able to be destination specific.

Proprietary communications are useful in two situations:

a. Where it is unnecessary to have standardized communications
b. Where it is important to communicate proprietary information

Much of the communications between nodes constructed by a single manufacturer do not require standardization. The information communicated is not generally useful to other devices on the network. In this situation the proprietary Parameter Group Numbers can be used.

When Parameter Group assignment is contemplated, proprietary and then PDU2 Format communications methods should be considered. If proprietary information is being communicated, the proprietary method should be used. If the information to be communicated is not of general interest, proprietary methods should be used. If the information is of general interest and does not require direction of the message to a particular node, a PDU2 Format assignment should be sought. Finally if the information is of general interest but requires direction to one or another devices, then destination specific addressing is needed and a PDU1 Format Parameter Group assignment should be sought.

5.9.3 Data Field Definition

Minimization of message overhead with CAN-based systems requires full use of the data field (all 8 bytes). Except in the case of very time-critical messages, related parameters should be grouped to fill the 8-byte data field. Following this principal conserves Parameter Group Numbers for future assignment. Strong justification is needed to allow definition of Parameter Groups that result in sparsely used data fields.

5.10 Transport Protocol Functions

Transport protocol functions are described as a part of the data link layer with the recognition that Transport protocol functionality is subdivided into two major functions: Message Packetization and Reassembly; and Connection Management. They are described in the following sections.

In the following paragraphs the term originator corresponds to the ECU or device that transmits the request-to-send message. The term responder corresponds to the ECU or device that transmits the clear-to-send message.

5.10.1 Packetization and Reassembly

Messages greater than 8 bytes in length are too large to fit into a single CAN Data Frame. They must therefore be broken into several smaller packets, and those packets transmitted in separate message frames. At the destination end, the individual message frames must be received and parsed and the original message reassembled from the received packets.

5.10.1.1 Message Packets

The CAN Data Frame includes an 8-byte data field. Because the individual packets which comprise a large message must be identified individually so that they may be reassembled correctly, the first byte of the data field is defined as the sequence number of the packet.

Individual message packets are assigned a sequence number of 1 to 255. This yields a maximum message size of (255 packets * 7 bytes/packet =) 1785 bytes.

5.10.1.2 Sequence Numbers

Sequence numbers are assigned to packets for transmission on the network during message packetization and then used on reception of packets to reassemble them back into a message.

Sequence numbers shall be assigned to individual packets beginning with one and continuing sequentially until the entire message has been packetized and transmitted. The packets shall be sent sequentially in ascending order starting with packet 1.

5.10.1.3 Packetization

A large message is defined as one whose data does not fit into the data field of a single CAN message frame (i.e. messages with a data field greater than 8 bytes).

For the purposes of this protocol, a large message is considered to be a Parameter Group that has associated with it a string of 9 or more bytes. The first Data Transfer Packet contains the sequence number one and the first 7 bytes of the string. The second 7 bytes are placed into another SAE

J1939/CAN data frame along with the sequence number 2, the third with sequence number 3, and so on until all the bytes in the original message have been placed into SAE J1939/CAN data frames and transmitted.

Each Data Transfer packet (other than the last packet in a transmission sequence) shall include 7 bytes of the original large message. The final packet includes a data field of 8 bytes: those being the sequence number of the packet and at least 1 byte of data related to the Parameter Group and then any remaining unused bytes set to "FF_{16}."

The time between packets for Multipackets Broadcast messages shall be 50 to 200 ms (reference Section 5.12.3). For multipacket messages directed to a specific destination, the originator maintains a maximum time between packets (where CTS allows more than one) of not more than 200 ms. Responders must be aware that the packets containing the data all have the same identifier.

5.10.1.4 Reassembly

Data packets are received sequentially. Each data packet of a multipacket message shall be assembled, in order of sequence number, into a single string of bytes. This string of bytes is passed to the application entity responsible for the large message.

5.10.2 Connection Management

Connection Management is concerned with the opening, use, and closure of virtual connections between nodes for destination specific transfers. A virtual connection in the SAE J1939 environment may be considered a temporary association of two nodes for the purpose of transferring a single large message that is described by a single Parameter Group Number (reference Appendix C—Figure C1 and Figure C2). In cases where the connection is from one to many there is no flow control or closure provided (see Appendix C—Figure C3).

5.10.2.1 Multipacket Broadcast

Large messages may be nondestination specific; that is, they may be broadcast messages. To broadcast a multipacket message, a node first transmits a Broadcast Announce Message (BAM). This message, which must be transmitted to the global destination address, constitutes a large message warning to the nodes on the network. The BAM message contains the Parameter Group Number of the large message to be broadcast, its size, and the number of packets into which it has been packeted. Nodes interested in the data are then required to allocate the resources necessary to receive and reassemble the message. The Data Transfer PGN (PGN = 60160) is then used to send the associated data.

5.10.2.2 Connection Initiation

A connection is initiated when a node transmits a Request to Send message to a destination address. The Request to Send message contains the size of the entire message in bytes, the number of separate messages in which it will be transferred, the Maximum Number of Packets that can be sent in response to one CTS, and the Parameter Group Number of the message being transported.

Upon receipt of a Request to Send message, a node may elect to accept the connection or to reject it. To accept the connection, the responder transmits a Clear to Send message. The Clear to Send message contains the number of packets it may accept and the sequence number of the first packet it is expecting. The responder must ensure that it has sufficient resources to handle the number of packets it is accepting delivery of. The sequence number of the packet, in the instance of a freshly opened connection, would be one. Note that the Clear to Send message may not include provision for all the component packets of the message. To reject the connection, the responder sends a Connection Abort message. The connection may be rejected for any reason although lack of resources, memory, etc., are likely to be the causes.

The connection is considered established for the originator (i.e. RTS device) when the originator receives a corresponding CTS from the responder (i.e. CTS Device). The connection is considered established for the responder when it has successfully transmitted its CTS message in response to an RTS. These definitions are used to determine when a Connection Abort must be sent to close a connection.

A responder should send a Connection Abort if it has looked at the RTS message and decided not to establish the connection. This allows the originator to move on to a new connection without having to wait for a timeout.

5.10.2.3 Data Transfer

Data transfer begins after the originator of the connection receives the Clear to Send message. An exception is if the Data Transfer was the result of the “Broadcast Announce Message.” In this case the Clear to Send message is not used. The PGN for Data Transfer is contained in the CAN identifier field of each packet. The first byte of the data field will contain the sequence number of the packet.

In the case of destination specific messages, the responder is responsible for coordinating flow control between the nodes. If the responder wants to stop data flow momentarily while a connection is open, it must use the Clear to Send message, setting the number of packets to send equal to zero. In the case where the flow must be stopped for some number of seconds, the responder must repeat the transmission of the Clear to Send message once per 0.5 S (T_h) to assure the originator the connection is not broken. All remaining bit fields are set to ones (Don't care).

5.10.2.4 Connection Closure

Two connection closure cases exist in the absence of errors. The first is when there is a global destination and the second is when there is a specific destination. In the case of the global destination no connection closure operation is performed beyond the reception of the data itself. See Sections 5.10.3 and 5.10.3.4. In the case of a destination specific transfer and upon receipt of the last packet in the message stream, the responder transmits an end-of-message acknowledgment to the originator of the message. This is the signal to the originator that the connection is considered closed by the responder. The "end of message ACK" closure is required to free the connection for subsequent use by other devices.

The "connection abort" message is not allowed to be used by responders in the case of a global destination (i.e. BAM). See Sections 5.10.3 and 5.10.3.4. In the case of a destination specific transfer, the originator or responder can, at any time, use the "connection abort" to terminate the connection. Refer to Section 5.10.2.2 for an explanation of when a connection is considered established for the originator and responder. If the responder should, for example, determine that there are no resources available for processing the message, it may simply abort the connection by issuing the connection abort message with abort reason 2 (see Table 7). Upon receipt of the connection abort message, any message packets already passed will be abandoned.

A failure of either node may also cause closure of a connection. For example: a gap of more than (T1) seconds after receipt of the last packet when more were expected (Clear_To_Send allowed more); a gap of more than (T2) seconds after a CTS was transmitted (originator failure); a lack of CTS or ACK for more than (T3) seconds after the last packet was transmitted (responder failure); a lack of a CTS for more than (T4) seconds after a CTS (0) message to "hold the connection open" will all cause a connection closure to occur. See Figure C1 in Appendix C and Section 5.12.3 regarding timeouts. When either the originator or responder decides to close out a connection for any reason including a timeout, it shall send a connection abort message with abort reason 3 from Table 7.

T_r = 200 ms
T_h = 500 ms
T1 = 750 ms
T2 = 1250 ms
T3 = 1250 ms
T4 = 1050 ms

With the definitions in this section and those in all sections under Section 5.10 the following observations can be made.

1. Connection closure for a broadcast announce message include the following.
   a. A connection is considered closed when the originator:
      i) Sends the last Data Transfer packet
   b. A connection is considered closed when the responder:
      i) Receives the last Data Transfer packet
      ii) Has a T1 connection timeout
2. Connection closure for Request-to-Send / Clear-to-Send messages include the following.
   a. A connection is considered closed when the originator:
      i) Receives the TP.CM_EndOfMsgACK at the completion of the data transfer for the entire PGN
      ii) Sends a Connection Abort for any reason (e.g. due to a T3 or T4 timeout)
      iii) Receives a Connection Abort

b. A connection is considered closed when the responder:

i) Sends the TP.CM_EndOfMsgACK at the completion of the data transfer for the entire PGN

ii) Receives a Connection Abort

iii) Sends a Connection Abort for any reason (e.g. including stopping the session early if desired, for a T1 or T2 connection timeout, etc.)

5.10.3 Transport Protocol—Connection Management Messages

This type of message is used to initiate and close connections and also to control flow. Transport protocol provides the following five transport protocol connection management messages: the Connection Mode Request To Send, the Connection Mode Clear To Send, the End of Message Acknowledgment, the Connection Abort, and the Broadcast Announce Message. The formats of these messages are shown in Figure 15 below in the Parameter Group definition for "Transport Protocol—Connection Management." Table D1 in Appendix D has the SPN assignments to parameters in this message.

Parameter Group Name:	**Transport Protocol—Connection Management (TP.CM)**
Definition:	Used for the transfer of Parameter Groups that have 9 bytes or more of data. A definition of each specific message defined as part of the transport protocol is contained in the Sections 5.10.3.1 through 5.10.3.5.
Transmission repetition rate:	Per the Parameter Group Number to be transferred
Data length:	8 bytes
Extended Data Page:	0
Data Page:	0
PDU Format:	236
PDU Specific:	Destination Address
Default priority:	7
Parameter Group Number:	60416 ($00EC00_{16}$)
Data ranges for parameters used by this Group Function:	
Control byte:	0-15, 18, 20-31, 33-254 are Reserved for SAE Assignment
Total Message Size, number of bytes:	9 to 1785 (2 bytes), zero to 8 and 1786 to 65535 not allowed
Total Number of Packets:	2 to 255 (1 byte), zero and 1 not allowed
Maximum Number of Packets:	2 to 255 (1byte), zero through 1 are not allowed
Number of Packets that can be sent:	0 to 255 (1 byte)
Next Packet Number to be Sent:	1 to 255 (1 byte), zero not allowed
Sequence Number:	1 to 255 (1 byte), zero not allowed

FIGURE 15 - FORMAT OF MESSAGES FOR TRANSPORT PROTOCOL

Connection Mode Request to Send (TP.CM_RTS): Destination Specific		
Byte:	1	Control byte = 16, Destination Specific Request_To_Send (RTS)
	2,3	Total message size, number of bytes
	4	Total number of packets
	5	Maximum number of packets that can be sent in response to one CTS. FF_{16} indicates that no limit exists for the originator.
	6-8	Parameter Group Number of the packeted message
	Byte 6	Parameter Group Number of requested information (8 LSB of parameter group number, bit 8 most significant)
	Byte 7	Parameter Group Number of requested information (2nd byte of parameter group number, bit 8 most significant)
	Byte 8	Parameter Group Number of requested information (8 MSBs of parameter group number, bit 8 most significant)
Connection Mode Clear to Send (TP.CM_CTS): Destination Specific		
Byte:	1	Control byte = 17, Destination Specific Clear_To_Send (CTS)
	2	Number of packets that can be sent. This value shall be no larger than the smaller of the two values in byte 4 and byte 5 of the RTS message.
	3	Next packet number to be sent
	4-5	Reserved for assignment by SAE, these bytes should be filled with FF_{16}
	6-8	Parameter Group Number of the packeted message
End of Message Acknowledgment (TP.CM_EndOfMsgACK): Destination Specific		
Byte:	1	Control byte = 19, End_of_Message Acknowledge
	2,3	Total message size, number of bytes
	4	Total number of packets
	5	Reserved for assignment by SAE, this byte should be filled with FF_{16}
	6-8	Parameter Group Number of the packeted message
Connection Abort (TP.Conn_Abort): Destination Specific		
Byte:	1	Control byte = 255, Connection Abort
	2	Connection Abort reason
	3-5	Reserved for assignment by SAE, these bytes should be filled with FF_{16}
	6-8	Parameter Group Number of the packeted message
Broadcast Announce Message (TP.CM_BAM): Global Destination		
Byte:	1	Control byte = 32, Broadcast Announce Message
	2,3	Total message size, number of bytes
	4	Total number of packets
	5	Reserved for assignment by SAE, this byte should be filled with FF_{16}
	6-8	Parameter Group Number of the packeted message

FIGURE 15 - FORMAT OF MESSAGES FOR TRANSPORT PROTOCOL (CONTINUED)

5.10.3.1 Connection Mode Request to Send (TP.CM_RTS)

The TP.CM_RTS message informs a node that another node on the network wishes to open a virtual connection with it. The TP.CM_RTS is a message with the source address field set to that of the originating node, the destination address field set to that of the intended recipient of a large message, and the remaining fields set appropriately for the Parameter Group Number being sent.

Byte 5 of this message allows the originator to limit the responder's number of packets specified in the Clear To Send message. See Figure C4 and Figure C5 in Appendix C. When the responder complies with this limit, it ensures that the originator can always retransmit packets that the responder may have not received for whatever reason.

If multiple RTSs are received from the same source address for the same PGN, then the most recent RTS shall be acted on and the previous RTSs will be abandoned. No abort message shall be sent for the abandoned RTSs in this specific case.

TP.CM_RTS is only transmitted by the originator.

5.10.3.2 Connection Mode Clear to Send (TP.CM_CTS)

The TP.CM_CTS message is used to respond to the Request To Send message. It informs the peer node that it is ready for a certain amount of large message data. The amount of large message data cleared to send shall be no larger than the smaller of the two values in byte 4 and byte 5 of the originator's TP.CM_RTS message.

If multiple CTSs are received after a connection is already established, then the connection shall be aborted. When the originator aborts the connection, it shall send the Connection Abort message with abort reason 4 from Table 7.

The responder will not send the next CTS until it has received the last data packet from the previous CTS or it has timed out. In the case of time out the responder has the choice whether to send a connection abort or to send a CTS.

The following cases exist when data transfer happens with errors:

- A missing or errant packet(s) is detected and the last packet is successfully received, then the responder will send a CTS requesting retransmission starting from the missing packet.
- Missing packet(s) including the last packet will lead to time out T1. In this case, the responder decides on sending a CTS or a connection abort with reason 4 from Table 7. See Figure C7.

When the CTS is used to request the retransmission of data packet(s), it is recommended not to use more than 2 retransmit requests. When this limit is reached, a connection abort with abort reason 5 from Table 7 shall be sent.

If a CTS is received while a connection is not established, it shall be ignored.

CTSs not only control the flow but also confirm correct receipt of any data packet prior to that CTS packet's number. Therefore if information for the previous CTS was corrupted, then a CTS for the corrupted information shall be sent before continuing on to the next sequential packets to be sent. Because of this requirement, the originator of a large message transmission may use byte 5 of the TP.CM_RTS message as a way to ensure the possibility of retransmission of a packet within the last set of packets cleared to send.

TP.CM_CTS is only transmitted by the responder.

5.10.3.3 End of Message Acknowledgment (TP.CM_EndOfMsgACK)

The TP.CM_EndOfMsgACK message is passed from the recipient of a large message to its originator indicating that the entire message was received and reassembled correctly. The responder can keep the connection open after the last Data Transfer of the session by not immediately sending the TP.CM_EndOfMsgACK. This allows the responder to get a packet resent if necessary.

If an End of Message Acknowledgment is received by the originator prior to the final Data Transfer, then the originator ignores it.

One End of Message Acknowledgment is sent to show the originator that the large message transfer has been received and assembled correctly.

TP.CM_EndOfMsgACK is only transmitted by the responder.

5.10.3.4 Connection Abort (TP.Conn_Abort)

The TP.Conn_Abort message is used by either node involved in a virtual connection to close the connection without completing the transfer of the message or to prevent a connection from being initialized.

Upon receipt of a Connection Mode Request To Send message, a node must determine if there are sufficient resources available to deal with the message for which this connection is sought. For example if the device must acquire memory from the system heap, it may not be able to claim enough to accept the entire message; or a device may simply be too occupied doing other things to expend processor cycles to handle a large message. In these cases a Connection Abort message may be sent even though the connection has not been established. This may be done in order to allow the originator to attempt another virtual connection without first having to wait for a timeout to occur.

When either the originator or responder decides to close out a connection for any reason, prior to completing the data transfer, including a timeout, it shall send a Connection Abort message with the appropriate Connection Abort reason. See Table 7 for the list of Connection Abort reasons.

TABLE 7 - CONNECTION ABORT REASON

Value	Description
1	Already in one or more connection managed sessions and cannot support another.
2	System resources were needed for another task so this connection managed session was terminated.
3	A timeout occurred and this is the connection abort to close the session.
4	CTS messages received when data transfer is in progress.
5	Maximum retransmit request limit reached
6–250	Reserved for SAE assignment.
251–255	Per J1939-71 definitions.

It is intended that the originator (i.e. the RTS node) should immediately stop transmitting after the reception of the Connection Abort message by the CAN protocol device. If this is not possible, the process to stop transmitting data packets shall take no more than 32 data packets and shall not exceed 50 ms. After sending or receiving a Connection Abort message, all related data packets received should be ignored. TP.Conn_Abort is transmitted by the originator or the responder.

5.10.3.5 Broadcast Announce Message (BAM)

The TP.CM_BAM is used to inform all the nodes of the network that a large message is about to be broadcast. It defines the parameter group and the number of bytes to be sent. After TP.CM_BAM is sent, the Data Transfer Messages are sent and they contain the packetized broadcast data.

TP.CM_BAM is only transmitted by the originator.

5.10.4 Transport Protocol—Data Transfer Message (TP.DT)

The TP.DT message is used to communicate the data associated with a Parameter Group. The TP.DT message is an individual packet of a multipacket message transfer (see Figure 16). For example if a large message had to be divided into 5 packets in order to be communicated, then there would be 5 TP.DT messages. Examples showing TP.DT messages being used can be seen in Appendix C. Table D1 in Appendix D has the SPN assignments to parameters in TP.DT shown in Figure 16.

TP.DT is only transmitted by the originator.

Parameter Group Name:		**Transport Protocol—Data Transfer (TP.DT)**
Definition:		Used for the transfer of data associated with Parameter Groups that have more than 8 bytes of data
Transmission repetition rate:		Per the Parameter Group to be transferred
Data length:		8 bytes
Extended Data Page:		0
Data Page:		0
PDU Format:		235
PDU specified field:		Destination address (Global (DA = 255) for TP.CM.BAM data transfers) (Global not allowed for RTS/CTS data transfers)
Default priority:		7 (Priority used regardless of the PGN being transported)
Parameter Group Number:		60160 ($00EB00_{16}$)
Data ranges for parameters used by this Group Function:		
Sequence Number:		1 to 255 (1 byte)
Byte:	1	Sequence Number
	2-8	Packetized Data (7 bytes). Note the last packet of a multipacket Parameter Group may require less than 8 data bytes. The extra bytes should be filled with FF_{16}.

FIGURE 16 - TRANSPORT PROTOCOL—DATA TRANSFER MESSAGE (TP.DT)

5.10.5 Connection Constraints

If a node cannot handle another session, then it should reject the connection initiations that are pursued by other nodes. An RTS for a different PGN from the same source address to the same destination as an existing session shall be rejected as well. In either case the newly requested session should be rejected by sending a Connection Abort with abort reason 1 from Table 7. This allows the device desiring a connection to move on to a new connection without having to wait for a timeout.

5.10.5.1 Number and Type of Connections a Node Must Support

Each node on the network can originate one destination specific connection transfer with a given destination at a time. This is due to the fact that the TP.DT only contains the source address and destination address and not the PGN of the data being transferred.

Only one multipacket BAM (i.e. global destination) can be sent from an originator at a given time. This is due to TP.DT not containing the actual PGN or a connection identifier. However, responders (i.e. receiving devices in this specific example) must recognize that multiple multipacket messages can be received, interspersed with one another, from different originators (i.e. source addresses).

A node must also be able to support one RTS/CTS session and one BAM session concurrently from the same source address. Therefore, the responder must use the destination address of the two transport protocol messages to keep them properly separated. One of the transport protocol messages has a global destination address and the other has a specific destination address. The destination address has to be used to distinguish the two because the TP.DT does not contain the actual PGN or a connection identifier.

Regardless of whether a node can support multiple simultaneous transport protocol sessions (RTS/CTS and/or BAM), it must assure that TP.DT messages from the same Source Address with different Destination Addresses can be differentiated. Receivers must use the Destination and Source Address to keep the data for the messages correct.

5.10.5.2 Intended Transport Protocol Use

Transport Protocol has been developed to provide a mechanism for transferring PGNs with 9 or more data bytes (see Section 5.2.7.2). A PGN defined as multipacket capable, having less than 9 data bytes to transfer in specific instance, shall be sent in a single CAN Data Frame with the DLC set to 8 (see Section 5.2.7.1).

5.10.5.3 Concurrent PGN Reception

It is possible that specific PGs may be sent in a nontransport protocol form when they are less than or equal to 8 bytes and then also be sent in transport protocol form when they are greater than 8 bytes. It is possible for these two forms of the same PG to be sent concurrently.

Note that a nontransport protocol form of a PGN is not considered to be a session so its being sent does not close out the transport protocol form of the same PGN.

5.11 PDU Processing Requirements

Processing of the PDUs requires specific procedures to be followed. Suggested sequence for interpreting PDUs is described in Appendix A. Appendix B shows example SAE J1939 message types and PDU formats being used.

Devices must be able to process data link messages fast enough to prevent losing messages when the data link is at 100 percent utilization. This also means that in low utilization situations, when there are back-to-back messages, each device must be able to process the messages fast enough not to lose messages due to their back-to-back nature. Processing the message fast enough does not mean that a response has to be immediately generated but that a new message must not overrun the previous messages.

5.12 Application Notes

5.12.1 High Data Rates

Data that is to be updated at a high rate and has tight latency requirements should, if possible, allow hardware-based message filtering to be used.

5.12.2 Request Scheduling

The scheduling of a request should be canceled if the information to be requested is received prior to the request being sent. That is, if the information is received 50 ms prior to request scheduling, the request shall not be issued. Parameter Groups should not be requested if they are recommended to be broadcast. Exceptions may arise when the recommended broadcast time exceeds a special case need.

5.12.3 Device Response Time and Timeout Defaults

All devices, when required to provide a response, must do so within 0.20s (T_r). All devices expecting a response must wait at least 1.25s (T3) before giving up or retrying. These times assure that any latencies due to bus access or message forwarding across bridges do not cause unwanted timeouts. Different time values can be used for specific applications when required. For instance, for high-speed control messages, a 20 ms response may be expected. Reordering any buffered messages may be necessary to accomplish the faster response. There is no restriction on minimum response time.

Time between packets of a multipacket message directed to a specific destination is 0 to 200 ms. This means that back-to-back messages can occur and they may contain the same identifier. The CTS mechanism can be used to assure a given time spacing between packets. The required time interval between packets of a Multipacket Broadcast message is 50 to 200 ms. A minimum time of 50 ms assures the responder has time to pull the message from the CAN hardware. The responder shall use a timeout of 250 ms (provides margin allowing for the maximum spacing of 200 ms).

Below is an example using the 1.25s (T3) timing.

a. Maximum forward delay time within a bridge is 50 ms (per J1939-31)
   Total number of bridges = 10 (i.e. 1 tractor + 5 trailers + 4 dollies = 10 bridges)
   Total network delay is 500 ms in one direction.
b. Number of request retries = 2 (3 requests total); this includes the situation where the CTS is used to request the retransmission of data packet(s). If the retransmit request limit is reached, then the connection abort shall be sent with abort reason 5 from Table 7.
c. 50 ms margin for timeouts

Appendix C—Figure C1 and Figure C3 have the timing requirements identified. In Figure C1 the time numbers are computed assuming the worst case number of bridges, 10 bridges. The timeout numbers for receivers are identified as a time value while transmitter requirements are specified as a less than or equal to time value. Note that an originator has transmitter and receiver requirements and that a responder has transmitter and receiver requirements.

5.12.4 Required Responses

A response is required for a global request from all devices that have the requested PG, even the requester. Acknowledgments are not allowed for global requests.

A device which uses a global Destination Address (DA = 255) for a request (e.g. “address request”) shall itself send a response if it has the data requested. This is a requirement because all devices are expected to respond. If the device issuing the request does not respond, then the other network devices may draw the wrong conclusion about the requested information.

5.12.5 Transmission of PGNs to Specific or Global Destinations

Most of the time it is desired to send periodically broadcasted PGNs to a global destination.

5.12.6 CTS Number of Packet Recommendation

During normal vehicle operation it is recommended that the maximum number of packets that can be sent per CTS be set to 16.

6. NOTES

6.1 Marginal Indicia

A change bar (l) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

PREPARED BY THE SAE TRUCK AND BUS CONTROL AND COMMUNICATIONS NETWORK COMMITTEE OF THE SAE TRUCK AND BUS ELECTRICAL AND ELECTRONICS STEERING COMMITTEE

APPENDIX A - SAE J1939 PDU PROCESSING—TYPICAL RECEIVE ROUTINE

A.1 RECEIVE INTERRUPT

When a message is received by the microprocessor via the CAN chip, several tests are performed in order to parse it and determine if and where it should be stored. J1939 processing of the CAN Data Frame will not occur if the Extended Data Page equals 1 and the Data Page equals 1, so the processing shown in Figure A1 does not apply. If the Extended Data Page equals 0 then the CAN Frame can be handled as shown in Figure A1. The three priority bits are used only for bus arbitration and therefore are not needed (used) by the receiving device. Note also that a given device may have more than one address if it performs multiple functions (see Figure A1).

```
IF PGN = REQUEST PGN AND THE DESTINATION IS SPECIFIC          ; specific request
THEN
  IF DA = ASSIGNED ADDRESS (destination)
  THEN
        SAVE 4 BYTE ID AND 3 DATA BYTES IN REQUEST QUEUE
IF PGN = REQUEST PGN AND THE DESTINATION IS GLOBAL            ; global request
THEN
  SAVE 4 BYTE ID AND 3 DATA BYTES IN REQUEST QUEUE

IF PF < 240
THEN
  IF DA = GLOBAL                                              ; PDU1 Format (DA = global)
  THEN
          USE JUMP TABLE FOR PGN VALUES OF INTEREST AND
          IF SA = ID OF SPECIAL INTEREST
          THEN
            SAVE 8 BYTES OF DATA IN DEDICATED BUFFER
          ELSE
            SAVE 12 BYTE MESSAGE (ID AND DATA) IN CIRCULAR QUEUE
  ELSE (DA = SPECIFIC)                                        ; PDU1 Format (DA = specific)
          USE JUMP TABLE FOR PGN VALUES OF INTEREST AND
          IF SA = ID OF SPECIAL INTEREST VALUES
          THEN
            SAVE 8 BYTES OF DATA IN DEDICATED BUFFER
          ELSE
            SAVE 12 BYTE MESSAGE (ID AND DATA) IN CIRCULAR QUEUE
IF PF > OR = 240
THEN                                                          ; PDU2 Format
  USE JUMP TABLE FOR PGN VALUES OF INTEREST AND
          IF SA = ID OF SPECIAL INTEREST
          THEN
            SAVE 8 BYTES OF DATA IN DEDICATED BUFFER
          ELSE
            SAVE 12 BYTE MESSAGE (ID AND DATA) IN CIRCULAR QUEUE
```

FIGURE A1 - TYPICAL RECEIVE ROUTINE

APPENDIX B - COMMUNICATION MESSAGE TYPES

B.1 EXAMPLE OF COMMUNICATION MESSAGE TYPES

This example shows how an engine would typically perform (see Figure B1):

1) BROADCAST/RESPONSE/ACK
   Send the engine serial number (Component ID Parameter Group Number = 65259 ($00FEEB_{16}$)).
2) DESTINATION SPECIFIC REQUEST (PGN 59904)
   Receive a specific request for the engine serial number. The message sent back is either a RESPONSE with the data, or a NACK. See item 4) below in the examples.

2A) GLOBAL REQUEST
   Receive a global request for the engine serial number. The message sent back is a RESPONSE from a specific device that has the data. Acknowledgments are not used on global requests.

3) COMMAND
   For some commands it may be desirable to have a specific acknowledgment that the task has been completed. When this is the case, a message can be sent back as either an ACK = COMMAND COMPLETE or a NACK = COMMAND NOT ABLE TO BE COMPLETED. The example in Figure B1 uses "CF" as the command that will be acknowledged with an ACK or NACK.
4) ACK
   Send the NACK message to indicate that the command or request could not be acted upon (invalid request). The NACK message contains the offending Parameter Group Number in the data field. If the Parameter Group Number in a COMMAND or REQUEST is not recognized by the destination (addressed device), it should still NACK. If the Parameter Group Number is recognized, but the parameter(s) are not available, a normal response is sent back but with the data value(s) set to 255.

	PF	PS(GE, DA)	SA	DATA
1) BROADCAST	254	235(GE)	000	236912
2) SPECIFIC REQUEST	234	000(DA)	003	PGN 65259
1) RESPONSE	254	235(GE)	000	236912
Or				
4) NACK	232	255(DA)	000	01,255,255,255,03,PGN 65259
2A) GLOBAL REQUEST	234	255(DA)	003	PGN 65259
1) RESPONSE	254	235(GE)	000	236912
3) COMMAND	CF	000(DA)	240	
1) ACK	232	255(DA)	000	00,255,255,255,240,PGN for CF
Or				
4) NACK	232	255(DA)	000	01,255,255,255,240,PGN for CF
Or OTHER[1]				

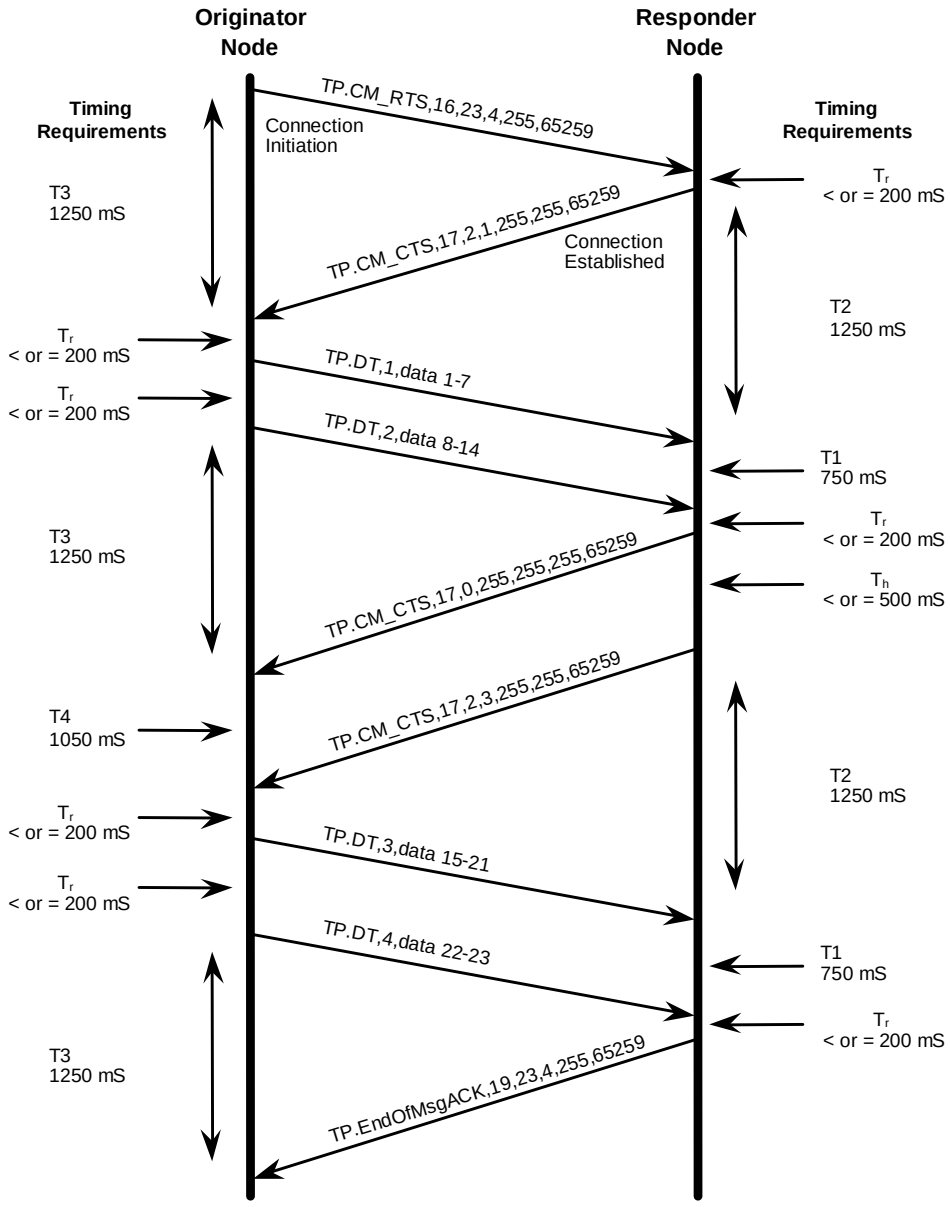
[1] COMMANDs must always have a mechanism for confirming that the action was successful or not. An ACK message is not required if another means is available. This helps to minimize bus traffic. For example a torque command to the engine can be confirmed by looking at the torque mode bits as well as the torque value coming from the engine.

FIGURE B1 - EXAMPLE OF COMMUNICATION MESSAGE TYPES

APPENDIX C - TRANSPORT PROTOCOL TRANSFER SEQUENCES

C.1 CONNECTION MODE DATA TRANSFER

Under normal circumstances, the flow model for data transfer follows Figure C1. The originator sends the TP.CM_RTS indicating there are 23 bytes in the packeted message, which will be transferred in four packets. The PGN for the data in the transfer is 65259, component identification.



NOTE—Timeouts (T1, T2, T3, T4) are described in Section 5.10.2.4.

FIGURE C1 - DATA TRANSFER WITHOUT ERRORS

The responder replies with a TP.CM_CTS indicating that it is ready to process two packets, beginning with packet 1.

The originating station passes the first two packets across the network using TP.DT. The responder station then issues a TP.CM_CTS indicating that it wants to hold the connection open but cannot receive any packets right now. A maximum of 500 ms later it must send another TP.CM_CTS message to hold the connection. In this example it sends another TP.CM_CTS indicating that it can take two more packets, beginning with packet 3. Once packets 3 and 4 have been transferred, the responder transmits a TP.EndOfMsgACK message indicating that all the packets expected were transmitted and that the connection is now considered closed. Note that packet 4 contains 2 bytes of valid data, bytes 22 and 23, the remaining data characters in this packet are transmitted as 255 (FF_{16}), data not available, such that the message is 8 bytes in length.

Message transfer in the event of an error on the link is shown in Figure C2. The TP.CM RTS is transferred and responded to properly; then data is lost during the data transfer phase.

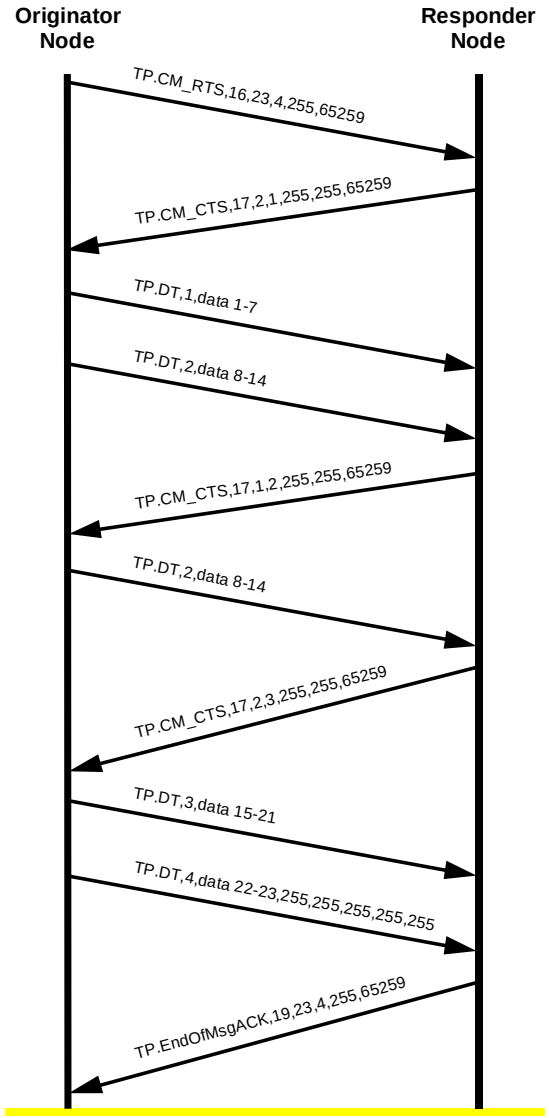


FIGURE C2 - DATA TRANSFER WITH ERRORS

In this situation, the request-to-send is sent in the same manner as the earlier example. The first two packets are transferred, but packet two was considered in error by the responder. The responder then transfers a TP.CM_CTS indicating that it wants a single packet, and that packet is packet 2. The originator complies, transferring packet 2. The responder then passes a CTS indicating it wants two packets, starting with packet 3. This TP.CM_CTS is the acknowledgment that packets 1 and 2 were received correctly. Once the last packet is received correctly, the responder passes a TP.EndOfMsgACK signaling that the entire message has been correctly received.

In the situation shown in Figure C3, a node indicates to the network that it is about to transfer a multipacket message utilizing the services of the transport protocol. In this example the PGN 65260, vehicle identification, is being broadcast to the network. The originating node first transmits a TP.CM_BAM (Broadcast Announce Message), followed by the data packets. No acknowledgment is performed by any of the responders (i.e. the receiving nodes in this specific example).

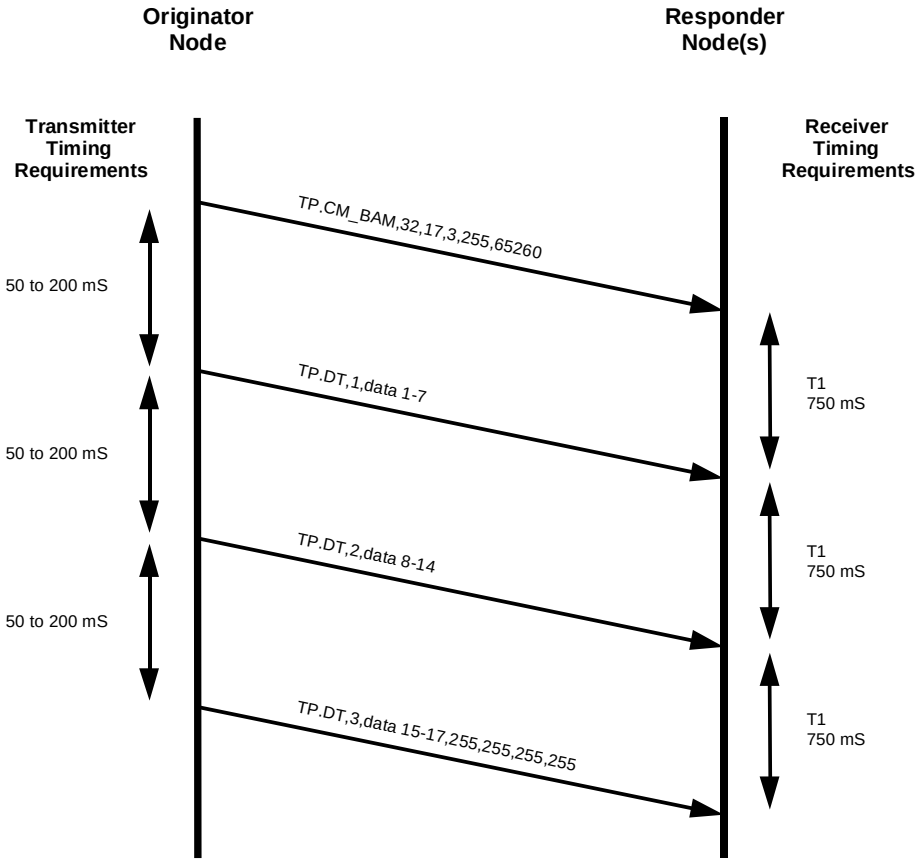


FIGURE C3 - BROADCAST DATA TRANSFER

In Figure C4 the originator uses the Maximum Number of Packets parameter to limit the number of packets the responder requests to be transferred. In this example both devices support the Maximum Number of Packets parameter.

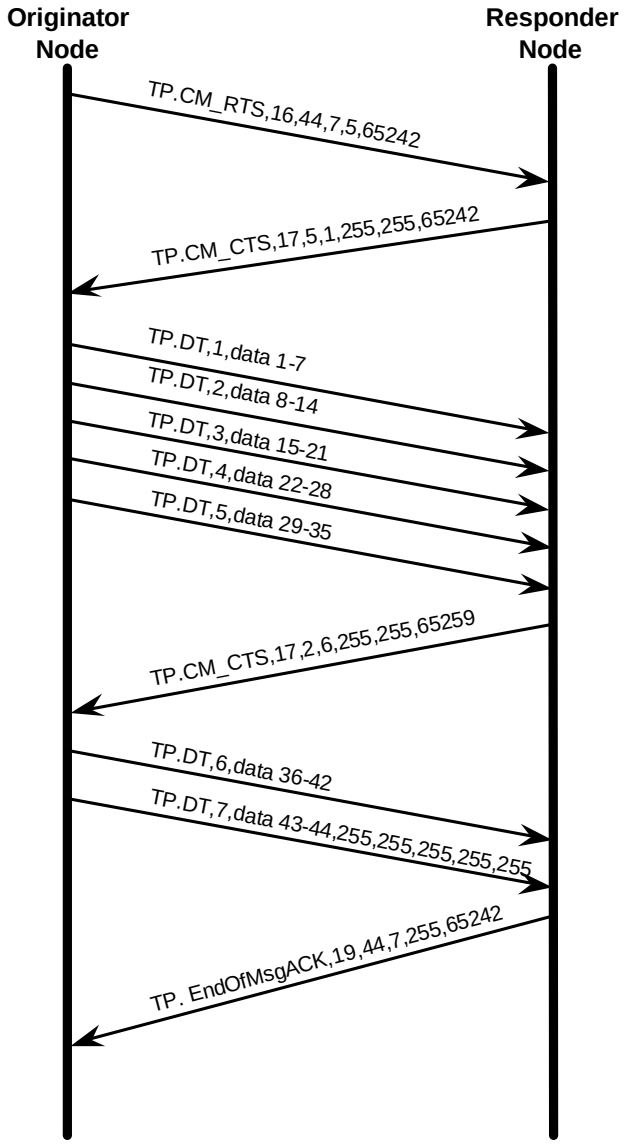


FIGURE C4 - DATA TRANSFER UTILIZING RTS MAXIMUM NUMBER OF PACKETS CAPABILITY

Figure C5 illustrates the situation where the originator supports the RTS parameter, “Maximum Number of Packets,” but the responder does not. In this situation the originator must comply with the responder’s CTS limits. In this example the originator would have to send seven data transfer packets even though it preferred to only send five at a time.

PRECAUTION: In this example if the responder sent a CTS for packet 1 after the data transfer of packet 7, the originator may have to recompute the information and therefore the second transmission of packet 1 may contain different data from the original packet 1 depending on the kind of data the PGN contains. For instance, PGN 65227 has dynamic data and might cause packet 1 to be different while a PGN such as 65242 has static data and would not cause packet 1 to be different on its second data transfer.

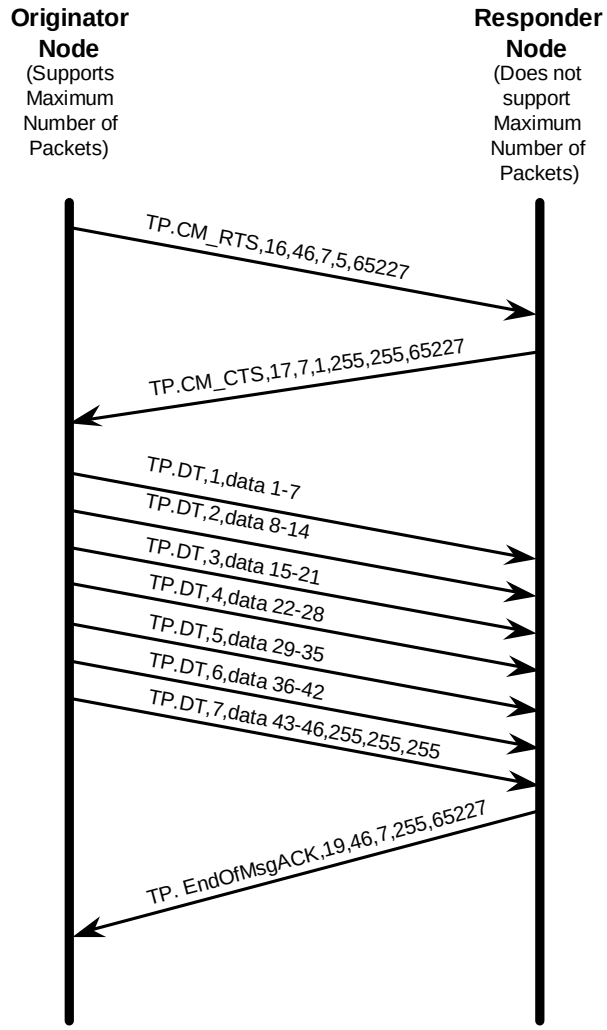


FIGURE C5 - DATA TRANSFER NOT ABLE TO UTILIZE RTS MAXIMUM NUMBER OF PACKETS CAPABILITY

Figure C56 shows the case where the last packet in a CTS group is missing. The responder sends a CTS in an attempt to avoid aborting the session.

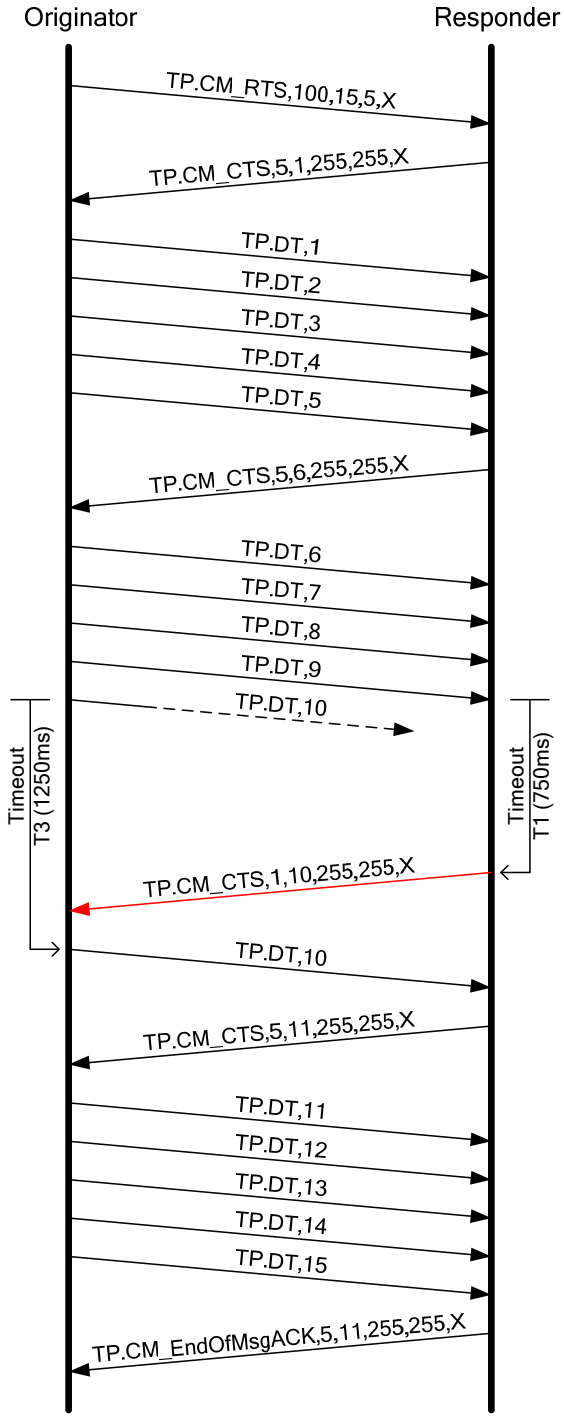


FIGURE C6 - RESPONDER SENDING CTS AFTER A TIME OUT TO PREVENT SESSION ABORT

In the case of CTS upon Time out, consider the following example:

In the slow originator example (left side), the originator is sending a 12 packet ‘CTS group’, but the responder's retransmit CTS arrives before the originator sends the last data packet. The originator aborts the session with abort reason 4. This is a possible example where the retransmit CTS does not avoid the session abort.

In the faster originator example (right side), the responder's retransmit CTS arrives after the originator sends the last data packet and before the T3 time out. This is the example where the originator is able to receive the CTS and avoid aborting the session.

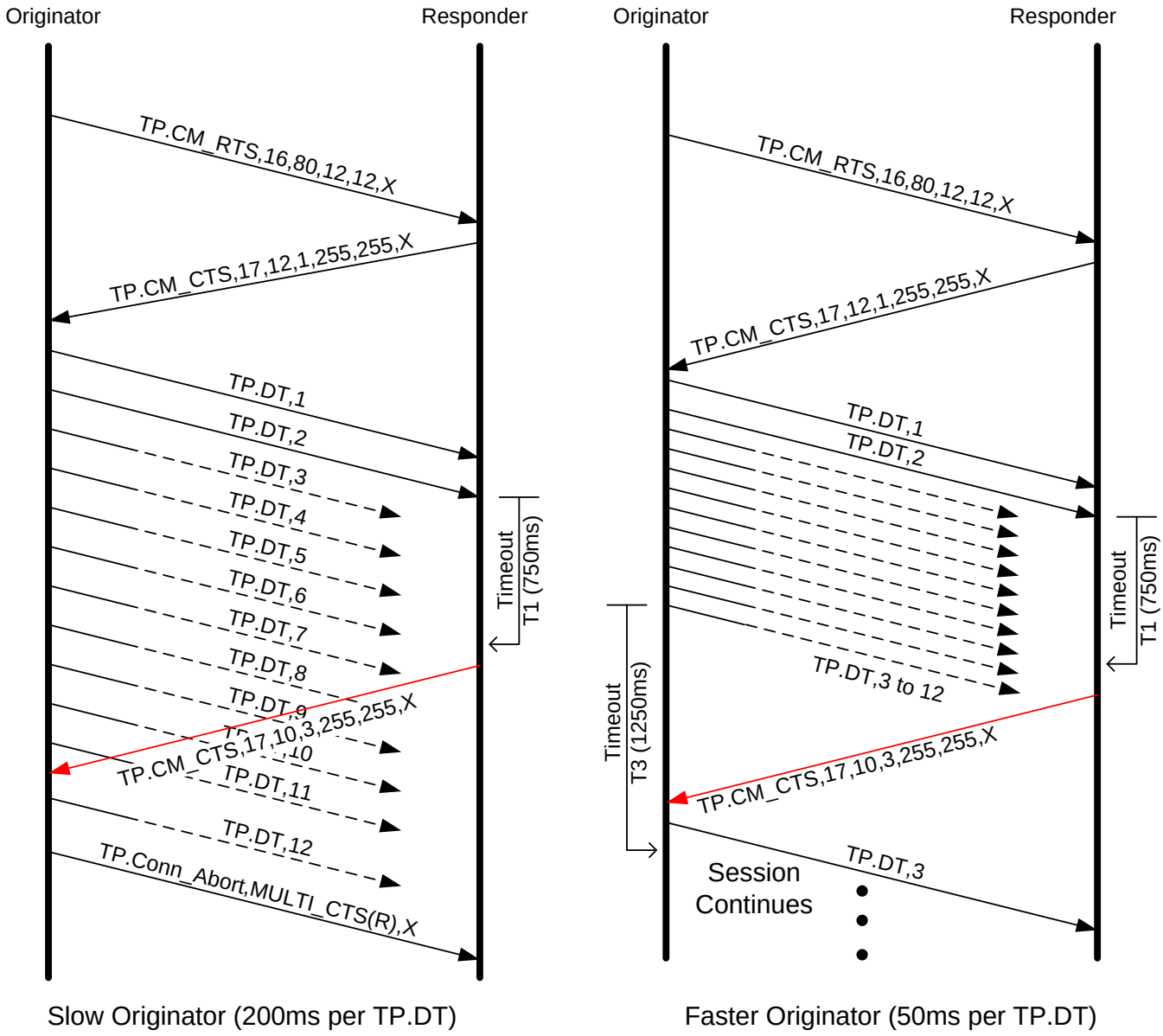


FIGURE C7 - EXAMPLES OF ORIGINATOR REACTION TO A RETRANSMIT CTS

APPENDIX D - ASSIGNMENTS OF SPNS FOR J1939-21

D.1 ASSIGNMENT OF SPNS FOR J1939-21

Table D1 shows the SPN assignments for J1939-21 items.

TABLE D1 - J1939-21 SPN USAGE

Row Number	PGN or Other Reference	SPN Name	SPN Number	SPN Bits[2]
1.	PGN 59904 Request	Parameter Group Number (RQST)	2540	24
2.	PGN 59392 Acknowledge	Control Byte (ACKM)	2541	8
3.		Group Function Value (ACK)	2542	8
4.		Address Acknowledged (ADD_ACK)	3290	8
5.		Parameter Group Number (ACK)	2543	24
6.		Group Function Value (NACK)	2544	8
7.		Address Negative Acknowledgement (ADD_NACK)	3291	8
8.		Parameter Group Number (NACK)	2545	24
9.		Group Function Value (NACK_AD)	2546	8
10.		Address Access Denied (ADD_AD)	3292	8
11.		Parameter Group Number (NACK_AD)	2547	24
12.		Group Function Value (NACK_Busy)	2548	8
13.		Address Busy (ADD_BUSY)	3293	8
14.		Parameter Group Number (NACK_Busy)	2549	24
15.	PGN 61184 Proprietary A	Manufacturer Specific Information (PropA_PDU1)	2550	64 to 14280
16.	PGN 126720 Proprietary A2	Manufacturer Specific Information (PropA2_PDU1)	3328	64 to 14280
17.	PGN 65280 to 65535 Proprietary B	Manufacturer Defined Usage (PropB_PDU2)	2551	64 to 14280
18.	PGN 51456 Request2	Parameter Group Number (RQST2)	2574	24
19.		Use Transfer Mode	2575	2
20.	PGN 51712 Transfer	Parameter Group Number of Requested Information (XFER)	2552	24
21.		Length of data for the reported PGN (XFER)	2553	8
22.		”Controller Application Identity” of the ECU for specific data subsets	2554	32
23.		Transfer Data	2555	72 to 14216

[2] The numbers in this column represent the size in bits of the data items identified by the SPN. The unit chosen is bits since some items are less than one byte in size.

Row Number	PGN or Other Reference	SPN Name	SPN Number	SPN Bits[2]
24.	PGN 60416 Transport Protocol–Connection Management	Control Byte (TP.CM)	2556	8
25.	TP.CM_RTS	Total Message size (TP.CM_RTS)	2557	16
26.		Total Number of Packets (TP.CM_RTS)	2558	8
27.		Maximum Number of Packets (TP.CM_RTS)	2559	8
28.		Parameter Group Number of packeted message (TP.CM_RTS)	2560	24
29.	TP.CM_CTS	Number of Packets that can be sent (TP.CM_CTS)	2561	8
30.		Next Packet Number to be sent (TP.CM_CTS)	2562	8
31.		Parameter Group Number of packeted message (TP.CM_CTS)	2563	24
32.	TP.CM_EndofMsgACK	Total Message size (TP.CM_EndofMsgACK)	2564	16
33.		Total Number of Packets (TP.CM_EndofMsgACK)	2565	8
34.		Parameter Group Number of packeted message (TP.CM_EndofMsgACK)	2566	24
35.	TP.Conn_Abort	Connection Abort Reason	2570	8
36.		Parameter Group Number of packeted message (TP.Conn_Abort)	2571	24
37.	TP.CM_BAM	Total Message Size (TP.CM_BAM)	2567	16
38.		Total Number of Packets (TP.CM_BAM)	2568	8
39.		Parameter Group Number of packeted message (TP.CM_BAM)	2569	24
40.	PGN 60160 Transport Protocol-Data Transfer	Sequence Number (TP.DT)	2572	8
41.		Packetized Data (TP.DT)	2573	72 to 14272